Family Size and Child Migration: Do Girls Face Greater Trade-Offs than Boys?*

Christine Ho[†] Yutao Wang[‡] Sharon Xuejing Zuo[§]

December 31, 2025

Abstract

Migrating with all family members is desirable but not always feasible in developing countries characterized with massive internal migration and significant rural-urban inequality. This study examines the causal impact of family size on children's rural-to-urban migration in China. Using a twin-based instrumental variable strategy, we first find that larger migrating families are more likely to leave children in the rural areas. More importantly, an additional sibling reduces the probability of a girl migrating by 12.9 percentage points, with a negligible effect on boys. The gender-differential effects are driven by a combination of son-biased fertility preferences and the diseconomies of scale in migration cost, rather than conventional son-biased allocative preferences.

Keywords: Child migration, family size, son preference, parental investment

JEL Codes: D13, J13, J16, O15, R23, R28

*We thank Janet Currie, Christian Dustmann, Tomoki Fujii, Rita Ginja, Naijia Guo, Rufei Guo, Seonghoon Kim, Hyejin Ku, Jing Li, Kathleen McGarry, Xin Meng, Sunha Myong, Saravana Ravindran, Christopher Taber, Alessandra Voena, Lingwei Wu, Yichong Zhang and attendees at the AASLE, AMES (China and Singapore), CCER Summer Institute, Exeter Diversity and Human Capital Workshop, GLO, MWIEDC, RES, and SEHO conferences, and Fordham, Fudan, Hamilton, Jinan, SMU, Southampton, and University of Macau seminars for insightful comments and suggestions. We are grateful to the National Health Commission of China for providing the China Migrants Dynamic Survey (CMDS) data. Ho acknowledges support from the Singapore MOE Tier 1(A) Grant 22-SOE-SMU-057 and Zuo acknowledges support from the National Natural Science Foundation of China (NSFC) (72003038 and 72121002) and the Innovative Research Groups Project of the NSFC (Grant 72121002). The authors declare that there are no conflict of interest with this research. All mistakes remain ours.

[†]Department of Social Policy, London School of Economics and School of Economics, Singapore Management University; ORCID: 0000-0002-2949-0184; Email: c.ho19@lse.ac.uk.

[‡]School of Economics, Singapore Management University; Email: ytwang.2020@phdecons.smu.edu.sg.

[§]School of Economics, Fudan University; Shanghai Institute of International Finance and Economics (SIIFE); Correspondence: sharonxzuo@fudan.edu.cn.

1 Introduction

Adult rural-to-urban migration is widely recognized as a driver of urbanization and economic development (Bau et al., 2025; Gai et al., 2025; Meng, 2012), representing a large-scale phenomenon in many developing economies. While the determinants of adult migration are well studied, far less is known about the decision to migrate with children, which is fundamentally shaped by the household structure. We thus address this gap by examining how family size affects child migration in China—a setting where son preferences and policy-induced migration restrictions are prevalent.

Rapid urbanization in developing economies is often accompanied by rural-urban inequality (UNESCO, 2019). China stands out for its particularly pronounced urban–rural divide. Urban areas offer superior public services—particularly education—which enhance human capital relative to rural schools. Consequently, whether a rural child migrates directly affects rural–urban inequality in human capital. This impact is amplified by scale: 46% of all children in China (138 million) have migrant parents (NBS et al., 2023), meaning that child migration decisions will profoundly affect the quality of the China's future labor force. Within this setting, the prevalence of son preferences and institutional migration restrictions creates a decision framework where household structure and resource constraints may shape child migration outcomes, with potentially gendered implications.

Motivated by these facts, this study addresses two central questions. First, how does family size affect child migration? Second, given the prevalence of son preferences, is the effect of family size gendered, and if so, what mechanism drives this differential effect?

By embedding son preferences and migration restrictions into the Beckerian quantity-quality trade-off theory, we first present a conceptual framework to illustrate the role of family size. On the one hand, driven by son-biased fertility preferences, parents are more likely to continue childbearing after a daughter is born, resulting in larger average sibship sizes for girls (Chae et al., 2025; Fan et al., 2018; Huang et al., 2024). This compositional difference means girls, on average, face stronger resource dilution and are thus less likely than boys to migrate. On the other hand, if migration restrictions create diseconomies of scale in migration cost, where the marginal cost rises with each additional child (a convex cost profile), then girls' overrepresentation in larger families further amplifies their disadvantage in migration opportunity, generating a gender-differential marginal effect of family size. This further widens the gender gap in resource dilution and, consequently, in child migration.

To test the empirical hypotheses, we use data from the China Migrants Dynamic Survey (CMDS), a large-scale nationally representative dataset on domestic migrants. In order to address the endogeneity of family size, we use the occurrence of twin births as an instrument to identify the causal effect. We find that the probability that a girl migrates decreases by 12.9 percentage points when she has one more sibling, whereas the negative effect of family size on boys is negligible. Moreover, conditional on family size, child gender per se does not affect the probability that a child migrates with parents, implying parents treat boys and girls equally in child migration for a given level of family size. This illustrates how equal treatment within households can produce unequal outcomes across gender through the channel of family size.

Robustness and sensitivity analyses confirm that the baseline findings hold across multiple alternative family structures and econometric specifications. Then, we conduct analyses on the heterogeneity of the results by child age and maternal education level. We find that family size effects on child migration are present only for primary school age children, but are negligible for other age groups. Moreover, the negative family size effects concentrate on children of mothers with an education attainment below the high school level.

Turning to the mechanism underlying the gendered effect of family size, we propose that it is primarily driven by the interaction of son-biased fertility with diseconomies of scale in migration costs. The data provide two pieces of evidence supporting the proposed mechanism. First, we find that the gender-differential effect of family size concentrates in destinations with strict migration policies, rather than in those with relaxed policies. Second, a non-linear specification reveals that the magnitude of the negative family size effect increases with family size, which aligns with a convex cost profile. Consequently, because son-biased fertility preferences place boys and girls at different points on the cost profile, they face different marginal costs, thereby generating gender-differential effect of family size.

Our study contributes to three strands of literature. First, we contribute to the quantity-quality (QQ) trade-off literature in two ways. On the one hand, we introduce a novel and direct measure of parental investment: child migration. Existing studies typically measure child quality using quality outcomes like educational attainment (see Guo et al. (2022) for a comprehensive review), but these conflate parental investment and child effort, and both respond to family size, making it difficult to isolate the parental decision. While a handful of studies use direct measures of parental investment, these are often limited to a single domain—educational spending (Chen, 2020; Dang and Rogers, 2016). Given that, a priori, the QQ effect may vary across investment types (Barcellos et al., 2014), we thus depart from previous studies by focusing on child migration, an unexplored investment that captures both time and monetary inputs.[1] On the other hand, we add to the literature on heterogeneous QQ effects by child gender. While some studies examine this gendered effect (e.g., Li et al. (2008)), the underlying mechanisms remain underexplored.[2] A notable exception is

[1]To the best of our knowledge, this is the first paper to study the effect of family size on the migration of preschool and primary school-age children. Existing literature is limited to older children (teenagers or adults), examining migration through the lens of their labor supply remittances (upstream transfers) (Bratti et al., 2020; Zhao and Zhong, 2019). Our context fundamentally differs, because the migration of young children is predominantly driven by parental investments (downstream transfers).

[2]It is noteworthy that our approach differs from another strand of literature that identifies a gendered effects by comparing same-sex children at a given parity based on whether they have an additional sister or brother (see e.g.,

Chae et al. (2025), who attribute the gendered QQ effect to unequal treatment within mixed-sex households, a family structure that grows more prevalent as family size increases. Nevertheless, we provide a new complementary explanation that links son-biased fertility preferences to the diseconomies of scale in migration costs, demonstrating a gender-differential effect even in the absence of within-household differential treatment.

Second, we add to the literature on how son preferences shape the gender gap in child migration rates. Prior research documents that boys are more likely than girls to migrate with parents in China, attributing this disparity largely to within-household differential treatment, where parents favor boys in resource allocation (Chen and Fu, 2023; Gao et al., 2025; Lin et al., 2021). However, a recent study indicates no significant parental preference for boys in child migration decisions in China (NBS et al., 2023). We are the first to reconcile these seemingly contradictory findings by demonstrating that the gender gap can arise even if parents treat boys and girls equally within households. Specifically, by appropriately dealing with the endogeneity of family size, we show that gender gap in the migration rate can be purely driven by a compositional effect in family size: due to son-biased fertility preferences, girls are more likely to belong larger families, where all children are less likely to migrate.

Third, we contribute to the emerging literature on how migration restrictions affect child migration (Chen and Fu, 2023; Dang et al., 2020; Gao et al., 2025; Huang and Zhang, 2023; UNESCO, 2019). Novelly, we provide evidence that migration restrictions can affect child migration rates and the associated gender gap through the channel of family size. This identifies family size as an unexamined mechanism connecting migration policies to the gender gap in child migration. Thus, our findings suggest that gender-neutral migration restriction policies can lead to gender-asymmetric migration outcome, which may contribute to widen gender gaps in human capital in the long run.

The remainder of this paper is structured as follows. Section 2 documents the institutional background and presents our conceptual framework. Section 3 introduces the data and presents summary statistics. Section 4 presents empirical strategies, our main results, heterogeneity analyses, and robustness checks. Section 5 investigates the mechanism, and Section 6 concludes.

2 Institutional Background and Conceptual Framework

2.1 Institutional Background

In 2020, there are around 376 million internal migrants in China, corresponding to nearly half of the world's population of intra-national migrants (NBS et al., 2023). This massive movement is predominantly characterized by rural-to-urban (RU) migration. RU migration used to be very restric-

Chen (2020) and Chen et al. (2019)).

tive prior to the 1980s due to the household registration system (i.e., *hukou*), whereby households registered in a rural area were prohibited to reside, work, or access education and health facilities in urban areas (An et al., 2024; Fan, 2019). In response to increased demand for labor, such restrictions were substantially relaxed since the 1980s, giving rise to huge waves of RU migration since the 1990s.

However, these rural migrant workers were more likely to work in the informal sector than urban natives, and they were often paid lower wages, even if they worked in the same industries or occupations as their native counterparts. In addition, they and their family members had limited access to social services, such as medical services and educational resources (Huang et al., 2024; Meng, 2012). These *hukou*-based barriers function as migration restrictions, and it is well established that *hukou* policies are closely linked to migration costs and thus to migration flows (An et al., 2024; Fan, 2019; Gai et al., 2025). As a result, rural migrant parents face significant challenges in bringing their children with them to urban cities even if they wanted to. According to the National Bureau of Statistics of China, about 71 millions or 24% of all children are migrant children in 2020.

It should be noted that although most urban residents were bound by the One-Child Policy (OCP), the majority rural mothers were allowed to have more than one child. Specifically, more than 80% of rural households have at least two children. According to Wang and Zhang (2018) and Zhang (2017), rural households faced more relaxed restrictions under the OCP such that China had a *de facto* two-tier policy that was more stringent for urban than rural couples as those in agriculture needed a larger family and many were so poor that they could not pay the one time fine anyway.[3]

2.2 Conceptual Framework

Benefits of Child Migration. There are at least two important human capital benefits for children to migrant with their parents. First, migrating with parents would involve an increased parental time investment. Many existing studies have found that time investment is more effective than monetary investment in producing child human capital (Del Boca et al., 2014; Yum, 2023). And as parental time investment in childhood is hard to be substituted by monetary investments, remittances from migrant parents may not be able to compensate the time investment losses experienced by their left-behind children. Second, children of RU migrants can have access to better educational and health resources compared to their rural counterparts, since urban areas tend to be more prosperous than rural areas (Chen and Fu, 2023; Wang and Zhang, 2018). The opportunity to gain access to better facilities coupled with the fact that RU migration tends to be long-term—at least 4–5 years for the majority of migrants (Wang et al., 2019)—can help improve migrant children's human capital (Lin

[3] Consistent with this, Guo et al. (2025) use census data to show that 80% of rural Han mothers had least two children in China (see Figure 2a in Guo et al. (2025)). Similarly, Guo et al. (2024) observe a comparable rate of 76% among rural households in another large-scale representative dataset.

et al., 2021; Wang and Chen, 2019).[4]

Hypotheses. Motivated by the insights of the quantity-quality trade-off theory (Becker and Lewis, 1973), the binding time and resource constraints faced by migrant families imply that larger families are less likely to bring their children to urban areas. Thus, it is straightforward for us to expect there is a negative association between family size and child migration. This delivers the following hypothesis.

Hypothesis 1 (Family size) *An increase in family size will decrease the probability that a child migrates with parents.*

Furthermore, we hypothesize that son preference combined with exogenous policy-driven migration restrictions plays a key role in parents' decision. First, theoretically, son preference manifests itself in two primary channels: a biased allocation of more resources to boys than to girls, as well as a greater desire for boys (Jayachandran and Kuziemko, 2011; Jensen, 2003). We refer to the former as the son-biased allocative preference and the latter as the son-biased fertility preference. Even without considering the migration restrictions, we should expect that:

Hypothesis 2 (Channel One) *If the son-biased allocative preference is present, we should expect that boys are more likely than girls to migrate with parents, conditional on family size.*

When the son-biased allocative preferences is present, parents would treat boys and girls unequally in migration decisions. If families have the same number of children but differ in their preference for allocating resources to enhance the quality of boys relative to girls, we should observe a gender difference in the probability of child migration, holding everything else equal.

On the other hand, the son-biased fertility preference indicates that couples will have a stronger preference to give additional birth if they have a girl rather than having a boy. As a result, girls will on average have more siblings than boys. If Hypothesis 1 holds, we will mechanically find that girls are less likely than boys to migrate with parents:

Hypothesis 3 (Channel Two) *If the son-biased fertility preference is present, we should expect that girls, on average, end up in larger families. As a result, boys are more likely than girls to migrate with parents. Furthermore, migration restrictions may exacerbate such gender gap in migration.*

In addition to son-biased fertility preference, another compelling feature of China's institution is the existence of strict migration restrictions. We further hypothesize that the marginal cost of

[4] Using data from the China Education Panel Study (2013–2014) we also find that migrant children tend to have higher standardized test scores and better health outcomes compared to left-behind children (Appendix Table A1).

bringing one additional child to urban areas will increase due to the *hukou* system, which generates an unequal distribution of social welfare between migrant workers and urban residents. This hypothesis is theoretically explained in the Model in Appendix A.1 and is supported by many anecdote facts (see Footnote A2). As a result, gender gap in child migration may be exacerbated.

Lastly, we would like to emphasize that Hypotheses 2 and 3 are not mutually exclusive. There can be a scenario where both channels operate simultaneously. To be specific, the son-preference fertility preference results in gender inequality in child migration between different sizes of families, while son-biased allocative preference will produce gender inequality within each size of family. Later, our empirical analysis will unravel one channel from the other.

3 Data and Summary Statistics

3.1 Data and Sample

We used the 2016 China Migrant Dynamic Survey (CMDS) for the main analysis.[5] CMDS is a nationally representative household survey, targeting all migrant workers since 2009, and has been widely used to study the migration and labor market outcomes of migrant families.[6]

The important reason for us to focus on the 2016 wave is because it contains rich information about each child's migration status as well as birth date, which provides the possibility for us to construct the dependent variable and the instrument. Our main sample is at the child-level, which includes 38,829 children. The sample comprises singleton children under age 12 from rural *hukou* families with at least two children. Since children aged 12 and below in the 2016 wave have been born after the nationwide ban on ultrasound prenatal screening in 2004 (Sun and Zhao, 2016), it will mitigate potential concerns about sex selection.[7] Later, we expand our sample to include all children under age 18. In addition, we follow the literature to focus on singleton children and drop twin children because twins are arguably different from singletons due to ambiguous birth orders and potentially different innate endowments (Bagger et al., 2021; Rosenzweig and Zhang, 2009). Instead, we retain a family-level indicator for a twin birth and the parity at which it occurred.[8] Lastly, the fact that about 80% of rural households have at least two children in China (Guo et al.,

[5]The survey was conducted by the Migrant Population Service Center of the National Health Commission. It adopts a stratified, multi-stage probability-proportional-to-size sampling method.

[6]For example, see An et al. (2024), Chen and Fu (2023), and Huang et al. (2024) among others.

[7]The Law of Population and Family Planning was implemented from September 2002 to April 2004, while the *Guan Ai Nv Hai Xing Dong* Program, which aimed to alleviate son-biased parental behaviors, was put in place in 2003. As part of our extensive sensitivity analyses in Section 4.4, we report results on the sample of first-born children whose sex ratio is biologically normal and tend to be as good as random (Ebenstein, 2010; Sun and Zhao, 2016).

[8]The twining rate in our sample—defined as the number of twin children over the total number of children in all households (including single child households)—is 1.71% (1,316 twin children out of 76,911 children), which is quite similar to that reported in previous studies (Li et al., 2008).

2024, 2025) provides us with a nice opportunity to use the occurrence of twin at the second parity as an instrument for family size.

Nevertheless, we use different criteria to construct analysis samples and carry out a rich set of robustness checks in Section 4.4. For example, we restrict the sample to first-born children. This will alleviate potential concerns on the endogeneity of child gender. We will discuss more details in Section 4.4.

3.2 Summary Statistics

Table 1 reports the summary statistics for the full sample of children, as well as for the subsamples of children from households with exactly two children and from households with three or more children. From the Column (1), for the full sample, 69% of children migrate with parents while approximately 50% of children are male and the average family size is 2.11 children. Children are on average 5.95 years old and 53% of them are of primary school age. Mothers and fathers are in their early thirties on average and 63-64% of children have parents with middle school education. The vast majority, 90%, of children have two migrant parents who migrated to the same destination city while 6% had grandparents (and/or great-grandparents) who migrated with the parents. 53% of parental migration is inter-province while 30% of parental migration is inter-prefecture. The average duration of migration of parents is around 5.22 years and 61% would be willing to settle in the destination city.

We would like to highlight two key descriptive findings. First, we find that girls are overrepresented in larger families. To be specific, from Columns (4) and (7) in Table 1, 51% (49%) of all children from families with two children are male (female) while 41% (59%) of all children from families with at least three children are male (female). Second, although the unconditional mean migration rate is identical for boys and girls, as shown in Columns (2) and (3) of Table 1, it may cover some interesting relationships between family size and child migration. For example, by comparing Columns (4) to (7), we find that families with three or more children exhibit significantly lower rates of child migration compared to two-child households. In addition, as shown in Columns (8) and (9), it is evident that the proportion of migration for boys is significantly higher than that for girls among children from households with three or more children.

4 Empirical Analyses

4.1 Empirical Strategy

Baseline OLS Specification. For child i in household j located in destination province p, we run the following baseline ordinary least squares (OLS) regression for the full analytical sample:

$$M_{ijp} = \beta_0 + \beta_1 FS_{jp} + \beta_2 Boy_{ijp} + \mathbf{X}'_{ijp}\mathbf{\Gamma} + \mu_p + \varepsilon_{ijp}, \tag{1}$$

where M_{ijp} is dummy variable that takes 1 if the child migrated with parents, and 0 otherwise. The family size, FS_{jp}, is defined as the number of children in a family. The error term ε_{ijp} is clustered at the household level to account for potential correlation across children in the same family.[9] The vector $\mathbf{X}_{ijp}$ contains controls at the child, parent, and household level. Specifically, we include child-specific controls such as an indicator for primary school age and relative birth order. We additionally include parental controls, such as maternal age at the second birth and indicator variables for parental education, and controls for household characteristics such as dummies for migration distance category.[10]

Our first point estimate of interest is β_1, which will describe the correlation between family size and child migration. In the next section, we will seriously discuss the endogeneity issue and argue that by using the occurrence of twin at the second birth parity as an instrument for family size, we are able to provide causal evidence of the effect of family size on child migration. This analysis will be used to test the Hypothesis 1.

In order to test Hypothesis 2, we include an indicator of whether the child is male, Boy_{ijp}, in the baseline regression. Conditional on family size, any statistically significant correlation between child gender and the probability of migration will suggest the existence of a son-biased allocative preference. For example, if the coefficient is positive, it indicates that boys are more likely than girls to migrate with parents.

Furthermore, in order to test the Hypothesis 3, we performed a subsample analysis by child gender and we are particularly interested in understanding whether family size will have gender differential impacts and if there are, whether such gender differences are statistically significant.

[9] Our results are robust to clustering at the prefecture level.

[10] Specifically, the vector $\mathbf{X}_{ijp}$ includes a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order (defined as $\frac{(BO_{ijp}-1)}{(FS_{jp}-1)}$, where BO_{ijp} is the raw birth order), maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years.

Instrumental Variable Strategy. A simple OLS regression cannot identify the causal effect of family size on child migration. For example, parents who have a preference for family cohesion may want to have fewer children and also be more willing to migrate with children. Thus, a negative association between the number of children and child migration may potentially be driven by unobserved family preferences, such that OLS estimates of family size effects would be biased downwards. Alternatively, parents who expect children to provide old age support may choose to have a larger family and also be more willing to migrate with children. In this case, the OLS estimates of the effects of family size would be biased upwards.

To identify the effects of family size on child migration, one needs an exogenous source of variation in fertility, which does not alter child migration status. The occurrence of twins is usually regarded as good as random and has often been used as an instrumental variable (IV) for family size in published literature (Bagger et al., 2021; Guo et al., 2025; Oliveira, 2016). We therefore use twinning at the second parity as an IV for the number of children and apply two-stage least squares (2SLS). For brevity, we only present the detailed 2SLS specifications for the full sample below, noting that the models for the boy and girl samples are estimated analogously.

The first stage of 2SLS for the full-sample baseline model is given by:

$$FS_{jp} = \alpha_0 + \alpha_1 Twin_{jp} + \alpha_2 Boy_{ijp} + \mathbf{X}'_{ijp}\mathbf{\Gamma} + \mu_p + \eta_{ijp}, \tag{2}$$

where $Twin_{jp}$ is an indicator that takes unity if the family had twins at the second parity and 0 otherwise. 0.41% of children in the analytic sample have second born twins in their family.[11]

The second stage uses the predicted $\widehat{FS}_{jp}$ from the first stage and is given by:

$$M_{ijp} = \beta_0 + \beta_1 \widehat{FS}_{jp} + \beta_2 Boy_{ijp} + \mathbf{X}'_{ijp}\mathbf{\Gamma} + \mu_p + \varepsilon_{ijp}. \tag{3}$$

Our twin IV strategy identifies the weighted average causal response from compliers, that is, families induced by twinning to switch from having fewer than n to at least $n+1$ children, for any $n = 2, 3, 4$ (Angrist et al., 2010). As documented above, those with a rural *hukou* were restricted only at the third birth. Thus, if all couples desire at least two children (see Section 3.2 for suggestive evidence)—as is often the case in rural areas (Guo et al., 2024)—then the increase in fertility due to twins at second birth would be orthogonal to unobserved parental preferences and constraints. Conversely, twins at higher parities may be systematically correlated with fertility as extra pregnancies are choices while firstborn twins would tend to have a lower proportion of compliers given that most rural households desire at least two children. In sensitivity analyses, we also use an indicator for the presence of twins at any parity as an alternative IV and find the results to be very similar.

[11] The twinning rate of second-born children is 0.40% as per the definition in Footnote 8.

Validity of the Twin Instrument. Although twinning is often regarded as good as random, there may still be some threats to identification. First, the presence of twins may cause parents to reallocate resources across siblings due to the shorter birth spacing and low birth endowment of twins (Rosenzweig and Zhang, 2009). If the presence of twins leads to greater (lower) resources being allocated towards singletons, then negative family size effects from 2SLS would be underestimated (overestimated). Second, unobserved maternal conditions may correlate with both the occurrence of twins and child outcomes (Bhalotra and Clarke, 2019, 2020). Given these concerns, we conduct balance checks by regressing our $Twin_{jp}$ indicator on several pre-determined household characteristics (Appendix Table A2). All coefficients are statistically insignificant at the 5% level. We further compare the socio-economic characteristics of families with and without second-born twins (Appendix Table A3). Besides maternal middle schooling, all other characteristics are relatively well-balanced, being either very close in magnitudes and/or not statistically significantly different between the two types of families. Thus, twin families seem to be similar in observable characteristics compared to non-twin families.

We additionally report the reduced-form OLS estimates of the IVs and conduct partial validity checks following Gai et al. (2025)—where we control for both $Twin_{jp}$ and FS_{jp}—in Table 4. When family size is not controlled for, there is a negative and significant association between $Twin_{jp}$ and the probability that a child migrates for the full and girl samples. However, the coefficient of $Twin_{jp}$ become statistically insignificant at the 5% level once family size is controlled for, suggesting that the correlations between family size and child migration have been absorbed by the family size covariate so that there is limited remaining associations between the IVs and child migration.[12] Hence, the exclusion restriction cannot be invalidated. To further mitigate concerns about the twin instruments, we derive bounds for family size effects using two separate methods. First, by assuming that selection on unobservables is proportional to selection on observables (Oster, 2019), and second, by relaxing the exclusion restriction (Nevo and Rosen, 2012).

Finally, similar to prior studies that used twinning as an instrument, we identify a local average treatment effect (LATE) from compliers (Oliveira, 2016). To shed light on the external validity of our estimates, we extrapolate to estimate average treatment effects (ATE) à la Brinch et al. (2017) in Section 4.4.

[12] Whereas twinning seems marginally significant for girls, a limitation of this test is that if the coefficient of the instrument is significant, then we cannot know whether this is due to the violation of exclusion restriction or due to the harmless correlation built by controlling for the collider variable (i.e., family size) when the exclusion restriction is valid.

4.2 Results

OLS Results. The baseline results for the full sample and the subsamples of boys and girls are presented in Table 3. From Column (1) there is a negative association of 2.5 percentage points between the number of children and the probability that a child migrates ($p < 0.01$). From Columns (4) and (5), having one more sibling is associated with a 3.5 percentage points decrease in the probability of migration for girls ($p < 0.01$) but with an insignificant 0.8 percentage points decrease for boys. The gender difference in the effect of family size is statistically significant at the 5% level. Nevertheless, we reckon that the OLS estimates may suffer from the endogeneity issues discussed above and thus focus on the 2SLS estimates.

First Stage of the 2SLS. We report the fist-stage results in Table 2. On the one hand, from Column (1), being a boy, rather than a girl, is associated with a reduction in family size, strongly supporting the presence of son-biased fertility preferences. In addition, in Columns (2) and (3), the estimated coefficient of the twinning dummy is statistically significantly larger for the boy sample than for the girl sample (coefficient equality test $p < 0.05$), thereby further providing compelling evidence for son-biased fertility preferences. Recall that all regressions are at the child level. Therefore, for all boys in the boy sample, their families must, by definition, have at least one non-twin boy. For these families, who would otherwise likely stop childbearing due to having sons, a twin birth is thus more likely to force a larger deviation from their intended family size. In contrast, families in the girl sample—those with at least one non-twin girl—are more likely to already be on a path to have more children in pursuit of sons, so a twin birth constitutes a smaller marginal deviation to their planned family size. Collectively, the above evidence strongly supports the presence of son-biased fertility preferences.

On the other hand, the first-stage Kleibergen-Paap F-statistics in all three 2SLS specifications of Table 3 are reasonably large and well above the rule-of-thumb thresholds, that are, 10 and 104.7 (Lee et al., 2022), suggesting that strong first stages.

Second Stage of the 2SLS. We now turn to the main 2SLS results from the second stage equation (3). First, from Column (2) of Table 3, having one more sibling is associated with a 7.9 percentage points decrease in the probability that a child migrates ($p < 0.05$). This is consistent with Hypothesis 1 and indicates the presence of a trade-off between family size and child migration. Compared to the 2SLS estimates, OLS estimates seem to be biased upwards (i.e., the magnitude of the negative family size effect is smaller for OLS than for 2SLS), which is consistent with the presence of unobserved factors that may potentially drive both family size and child migration. As discussed above, one such factor may be parental desire to receive old age support from children, which may incentivize parents to have more children and be more likely to migrate with children.

Second, we examine whether gender *per se* matters for child migration. From Columns (1) and (2) of Table 3, there is no economically or statistically significant association between child gender and the probability that the child migrates. Specifically, the marginal effect of being a boy rather than a girl is neither statistically significantly different from zero at the 10% level in the OLS nor 2SLS regression. Therefore, conditional on family size, we find that boys and girls are equally likely to migrate with parents, providing no support for Hypothesis 2.[13]

Lastly, we focus on the gender-differential family size effect. From Columns (5) and (6) of Table 3, a one-unit increase in family size leads to a 12.9 percentage points decrease in the probability that a girl migrates ($p < 0.05$). Conversely, a one-unit increase in family size has a statistically insignificant negative effect on boys, with a very small magnitude. The gender difference in family size effects on child migration is 11.8 percentage points ($p < 0.10$). Combining first-stage evidence of son-biased fertility preferences with these results—which align with Hypothesis 3 but contradict Hypothesis 2—we find that the lower migration rate of girls is driven by son-biased fertility preferences rather than son-biased allocative preferences, an "equal treatment, unequal outcome" scenario. Notably, the effect of family size itself is gender-differential, further widening the gender gap in migration rate. Given the nature of child migration, we hypothesize that this gender-differential marginal effect stems from the interaction of son-biased fertility preferences with migration restrictions. Later, we revisit and formally examine this mechanism in Section 5.

4.3 Heterogeneity Analyses

In this subsection, we explore the heterogeneity of the gendered family size effects by child age groups, as well as maternal education attainment. First, we explore the heterogeneity by child age. Given that the main driver of parental migration is the potential to earn higher income, parents may be more willing to bring school age children who can spend most of the working week at school rather than pre-school age children who require more substantial child care. Moreover, if child migration is driven by parents' desire to invest in their children's human capital, then one would also expect school age children to be more likely to migrate compared to pre-school age children.[14] Thus, we conjecture that the gender differences in the effect of family size on child migration may be more pertinent for primary school age children.

Second, we further examine the heterogeneity by maternal education. Given the role of son preference in determining fertility, as well as its asymmetry between spouses, we expect the gendered family-size effect to be heterogeneous by maternal education. This is because a mother's

[13]We also follow Jensen (2003) to obtain a within-family estimator of child gender dummy by employing a family fixed effect (FFE) regression, where family size is absorbed by the FFE. From the Column (1) in Appendix Table A7, we do not find gender-per-se effect either, providing additional evidence *against* Hypothesis 2.

[14]Aligning with this, and while not reported in our baseline 2SLS results, we indeed find that primary school age children are 3.2 percentage points more likely to migrate compared to preschool age counterparts.

bargaining power, which can be proxied by her education relative to her spouse's, critically shapes how these preferences translate into parental investment decisions.

By Child Age. We now turn to examining the heterogeneity of child age on the gendered family size effect. We extend our sample to include older children and conduct separate analyses on the subsamples of children aged 0–5 (pre-school age), 6–12 (primary school age), and 13–18 (middle school age or above) in Table 5. From Columns (4) and (6), there are clear family size effects on child migration, especially for girls, among children of primary school age (6–12). Among children aged 6 to 12, the differential magnitude of family size effects for boys and girls, though a formal test of equality is underpowered due to the small sample size, is particularly consistent to the baseline gendered patterns. Conversely, there is no statistically significant evidence of family size effects among either preschoolers (0–5) or older cohorts (13–18). Therefore, the analyses from the extended sample confirm that the negative family size effects on children' migration are the most relevant for primary school age children. As discussed above, migrant mothers tend to lack access to affordable child care, so that migrant parents may be more reluctant to bring pre-school age children with them. Conversely, the migration of older children aged above the legal working age of 16 may also be clouded by their own labor supply decision.[15]

By Maternal Educational Attainment. To examine heterogeneity by maternal education, we split the sample by whether the mother's education attainment is high school or above. Table 6 shows the gendered family size effect is concentrated among children of less-educated mothers. This aligns with a bargaining-power narrative: lower maternal education correlates with weaker intra-household bargaining power. Given that mothers typically exhibit weaker son preference than fathers in patriarchal societies (Dizon-Ross and Jayachandran, 2023; Qian, 2008), households where the mother has less say are more likely to follow the father's son-biased fertility preferences. Consequently, girls in these families are more systematically overrepresented in larger households.

We further validate this mechanism using paternal education. The effect is concentrated in families where the father is highly educated—a pattern consistent with "marrying-up," where highly educated fathers are often paired with less-educated mothers, again rendering the father's son preference to dominate fertility decisions. The convergence of results across both maternal and paternal education subsamples reinforces that son-biased fertility—and thus the gendered family size effect—is stronger when the father's preference dominates intra-household bargaining.

[15] We note that the migration decision for older children may be confounded by their own labor supply decisions or by incentives to take the National College Entrance Exam (NCEE). The NCEE can only be taken in the province of origin and since different provinces use different syllabuses, migrant children have greater incentives to enroll in middle or high school in the region of origin of their *hukou* (Chen and Feng, 2017; UNESCO, 2019).

4.4 Robustness Checks

Our findings and inferences on gendered family size effects hold in extensive sensitivity checks. We first assess the robustness of our results through four alternative family structures. In addition, we summarize the methodological sensitivity checks and provide more details in Appendix B.

Including One-Child Families. As discussed in Section 3.1, it is common to exclude single-child households in twin studies (Black et al., 2005). Moreover, the vast majority of rural households have at least two children in China (Guo et al., 2024, 2025). Nevertheless, we also perform robustness checks by extending the sample to include only-children and use the presence of first-born twins as the instrument in this case. From Table A4, the findings and inferences on gendered family size effects are robust to this extension.

Families with Completed Fertility. As the CMDS contains some young respondents, some may not have yet completed their fertility decisions. We thus perform sensitivity analyses by restricting the sample to children in families where the first or youngest child is aged strictly above six since most Chinese mothers would have completed their fertility by that time (Huang et al., 2021). As can be seen from Appendix Table A5, the results are robust to this sample restriction.

Firstborn Children. As per Angrist et al. (2010), the estimated effects for lower parity children may be more precise than for higher parity children, because the outcomes of the last-born child come from an endogenously selected sample if fertility is endogenous. We thus follow Black et al. (2005) and employ a sample truncation specification to re-estimate the family size effects. In our context, we re-estimate the 2SLS models on the sample of firstborn children. More importantly, this set of sensitivity analyses also helps further mitigate potential concerns on the endogeneity of child gender since firstborn children's gender are generally believed to be random (Ebenstein, 2010; Huang et al., 2024; Sun and Zhao, 2016). From Columns (1)–(3) of Appendix Table A6, the family size effects are similar to those from the main models in Columns (2), (5), and (6) of Table 3.

Children Born Outside of the Destination City. Our measure of child migration, M_{ijp}, captures both children who migrated by travelling from a rural to an urban area with the parents and children who were born to migrant parents in the destination cities (i.e., children born before and after the parent migrated to the city). Note that although children may be born in cities, migrant parents would still need to register the children in the village of origin such that the child would have the same rural *hukou* as the parents (An et al., 2024). We refine the sample by considering only children who were born outside the destination city (68.4% of children in the main sample). From Columns

(4)–(6) of Appendix Table A6, the family size effects are once again comparable to those from the main models in Columns (2), (5), and (6) of Table 3.

Mixed-Sex Households. Our baseline 2SLS results—showing no gender-per-se effect—exploit between-household variation. To further demonstrate its robustness, we conduct two checks. First, we conduct a compositional analysis of the relevant subsample. We note that while 63.1% of children in our sample are from mixed-sex families where within-household differential treatment could plausibly occur, only a small fraction (6.6%) of families are in the partial-migration scenario, where some children migrate while others are left behind. Second, we directly exploit within-household variation in child gender by estimating a family fixed effects (FFE) regression on mixed-sex families (see also Footnote 13). The coefficient of the boy dummy is statistically insignificant and close to zero. Taken together, these results confirm that the absence of a gender-per-se effect is robust even in households where within-family variation in child gender exists.

Econometrics Concerns. We also demonstrate the methodological robustness of our baseline results through following sensitivity checks:

(1) *Bounding family size effects*: We bound family size effects in two ways. First, we follow Oster (2019) to bound the family size effects using selection on observed variables as a guide to selection on unobserved variables in the OLS model (1). Second, we relax the exclusion restriction in the 2SLS model (3) and estimate bounds on family size effects following the method proposed by Nevo and Rosen (2012). Figure 1 reports the estimated bounds and 95% confidence intervals for family size effects. As can be seen from the figure, our results and inference that family size effects matter for girls but not for boys remain robust.

(2) *Alternative specifications*: We perform two additional robustness checks. First, we use a two-step approach to address the concern that birth order configuration may be jointly determined with family size. In step one, FFE models identify birth order effects and in step two, 2SLS uses the predicted migration outcome net of birth order effects from step one, with block-bootstrapped standard errors (Bagger et al., 2021; Bratti et al., 2020). An important by-product of the two-step approach is that the FFE regression can help us identify the gender-per-se effect via within-household variation, further lending support to our baseline results. Second, we estimate probit models with control functions to address the discrete nature of child migration (Wooldridge, 2010). Appendix Table A7 shows that both specifications yield results similar to our main estimates, confirming our findings are robust to birth order confounding and functional form assumptions.

(3) *Moving beyond LATE and estimating ATE*: Finally, as mentioned above, our estimates recover the LATE from the relatively small set of compliers. As a guide to the relevance of our results

to a more general population, we extrapolate from the LATE to estimate ATE (Brinch et al., 2017). The ATE for boys is negligible (-0.006, $p > 0.1$) while the ATE for girls is -0.111 ($p < 0.05$), which is once again consistent with gendered family size effects. See Appendix B for details.

5 Mechanism Discussions

First, we do not find evidence to support the son-biased allocative preference, as discussed in Section 4.2. Instead, we find that the overrepresentation of girls in larger families, which results from the son-biased fertility preference, can explain the average lower migration rate for girls. However, this does not necessarily suggest that there are gender-differential effects of family size on child migration (see Figure A1 for graphical illustrations).[16] In other words, the result of significant gender differences in the family size effect presented by subsample analysis indicates that there should be other factors that may influence child migration decisions.

In this section, we explore the importance of migration restrictions on parents' decisions and how it further drives the gender-differential effect.[17] There are at least two reasons for us to focus on China's internal migration restriction policies. First, it is reasonable to predict that parental preferences and migration policies will jointly determine child migration in the context of China. As discussed in Section 2, due to migration restrictions, urban-rural areas have been segregated for decades. It is expensive for migrant parents to bring children with them. Second, more importantly, prior literature indicates that migration costs often exhibit diseconomies of scale (Davis et al., 2021; Mincer, 1978), with a convex relationship between total migration cost and family size. Thus, we propose that the marginal cost of migrating a child progressively increases with family size. In addition to the theoretical model presented in Appendix A.1, many existing studies have documented the facts that shed light on the increasing rate of marginal cost. For example, a seat in a public school and a fee reduction for one child may be secured through networks (Peng, 2019), but securing similar benefits for additional children is considerably harder due to diminishing social network assistance (Epstein, 2008). Consequently, parents are often forced to enroll subsequent

[16] Specifically, as shown in Figure A1a, girls' overrepresentation in larger families only results in that girls are concentrated at the upper end and boys at the lower end of the family size distribution, as well as lower migration rates for girls. However, it does not, on its own, produce a gender-differential effect under a linear cost structure. In this scenario, the marginal effect of family size remains equal for boys and girls.

[17] Chae et al. (2025) provide an alternative mechanism for the gendered family size effect. They find that family size has a more negative effect on girls' test scores than boys', attributing this to the rising prevalence of mixed-sex families with larger family size and son-biased allocative preferences within them. Thus, a girl in a newly mixed-sex family suffers both resource dilution and differential treatment, whereas a boy in the same situation may receive a relative boost. However, this narrative is unlikely to explain our context. As shown in Section 4.4, we find no evidence of son-biased allocative preferences for mixed-sex households. Consequently, the gender-differential family-size effect does not operate through this alternative channel.

children in costlier private for-profit schools, thereby raising the marginal cost of migration (Huang et al., 2024; Wang et al., 2017).

5.1 Destination *Hukou* Policy Stringency

Following the existing literature, we use the *hukou* policy as a measure of migration restrictions (An et al., 2024; Chen and Fu, 2023; Fan, 2019). It is evident that easing *hukou* restrictions led to a significant reduction in migration costs (Gai et al., 2025) and a substantial influx of migrants (An et al., 2024; Fan, 2019), confirming a tight link between *hukou* policy and migration restrictions in China. Further, Chen and Fu (2023) demonstrate that cities aiming to curb migration population often implement stringent barriers to the access of public education for children with non-local *hukou*. This evidence collectively establishes *hukou* policy as a reliable proxy for migration restrictions.

To begin with, we directly test the proposed mechanism of migration restrictions by examining whether family size effects are stronger for children with parents migrating to destination prefectures with strict *hukou* policies. To do so, we compile the composite *hukou* index constructed by Zhang and Lu (2019) for 2014–2016, which has been extensively adopted in recent studies (see e.g., An et al. (2024) among others).[18] The index comprises of four dimensions that measure the difficulty in obtaining a local *hukou* by investment in a prefectural city, housing property purchase, high-end employment, and ordinary employment. Thus, the composite index integrating these four components measures the aggregate stringency of local *hukou* qualifications in each prefectural city.[19] Consistent with prior literature, we observe substantial heterogeneity in *hukou* policies across prefectures with the index ranging from 0.275 to 2.628, where a higher value reflects greater difficulty for migrants to obtain a local *hukou*.

We utilize the median value of the composite measure as the cutoff to construct an indicator for whether the destination prefecture has strict *hukou* policies ($Strict_{jp}$). Then, we estimate an augmented 2SLS model of equation (1), which includes a $Strict_{jp}$ dummy and its interaction with family size ($FS_{jp} \times Strict_{jp}$):

$$M_{ijp} = \gamma_0 + \gamma_1 FS_{jp} + \gamma_2 FS_{jp} \times Strict_{jp} + \gamma_3 Strict_{jp} + \mathbf{X}'_{ijp}\mathbf{\Gamma} + \mu_p + \varepsilon_{ijp}, \quad (4)$$

where the marginal effect (ME) of family size for relaxed destination prefectures is given by γ_1 while the ME of family size for strict destinations is given by $(\gamma_1 + \gamma_2)$. As shown in Table 7,

[18] To further show the robustness of the mechanism, we also experiment with an alternative index measuring the degree of a migrant's job stability and the prospect of long-term settlement at a particular destination, known as *hukou* policy liberalization annual index (Fan, 2019; Gai et al., 2025). The resulting 2SLS estimates remain robust.

[19] Zhang and Lu (2019) construct the index through: (i) categorizing urban *hukou* policy documents, (ii) extracting key information on household registration requirements including investment, housing purchase, talent recruitment, and employment, (iii) implementing standardized scoring of policy stringency, and (iv) integrating components using a Projection Pursuit Model.

from Columns (2) and (3), The main effect of family size is insignificantly negative with similar magnitude for boys and girls. A formal test further shows that the statistical equality of the family size effect for boys and girls in relaxed-policy destinations ($p > 0.10$), indicating an absence of significant family-size effects under such relaxed *hukou* policies.

However, the interaction with the strict-policy dummy is significant only for girls, indicating a differential effect of family size across policy stringency specifically for them. For boys, this differential effect in strict destinations remains statistically indistinguishable from zero. By combining the main and interaction effects, we compute the ME for strict prefectures. This calculation reveals clear family-size effects for girls in prefectures with strict *hukou* policies, with a negligible effect for boys. A formal test confirms the presence of a statistically significant gender gap in the family size effect under strict policies ($p < 0.05$). Thus, the patterns observed in strict-policy destinations align with our baseline results, whereas those in relaxed-policy areas do not, implying that the gendered family size effect is driven by migration restrictions.

5.2 A Convex Profile of the Marginal Effect of Family Size

As mentioned above, the literature suggests the presence of diseconomies of scale in migration costs (Davis et al., 2021; Mincer, 1978). In our context, migration restrictions impose the cost of child migration is convex in family size, resulting in a convexly increasing negative effect of family size on migration probability.

To provide suggestive evidence, we estimate a non-linear model of family size effect, where both birth order and family size profiles are non-linear. Specifically, we employ 2SLS to estimate a revised version of equation (1):

$$M_{ijp} = \beta_0 + \beta_1 Boy_{ijp} + \phi_3 \mathbb{I}\left\{FS_{jp} = 3\right\} + \phi_{4^+} \mathbb{I}\left\{FS_{jp} \geq 4\right\} + \mathbf{X}'_{ijp}\boldsymbol{\Gamma} + \mu_p + \varepsilon_{ijp}, \tag{5}$$

where we includes two indicators for families with three and for those with four or more children, while using families with exactly two children as the reference category.[20] Then, we use twinning at second (third or higher) parity as the instrument for the indicator $\mathbb{I}\left\{FS_{jp} = 3\right\}$ (indicator $\mathbb{I}\left\{FS_{jp} \geq 4\right\}$) (see Appendix C for details). Table A9 presents the results. Specifically, from the Column (1), the coefficients for families with three children ($\hat{\phi}_3 = -0.086$, $p = 0.033$) and four or more children ($\hat{\phi}_{4^+} = -0.22$, $p = 0.052$) are significantly negative, and follow a monotonic pattern.[21] Therefore, compared to children from two-child families, children from larger families face

[20] Note that our sample is restricted to families with at least two children, with a maximum family size of five. Due to the small proportion of households with five children, we combine them with households of four children.

[21] This convex pattern persists in the boy and girl subsamples, with a consistently larger adverse effect observed for girls. The gender difference in magnitude is evident across all parities, though underpowered due to the rareness of twins at higher parities.

an increasingly greater adverse family size effect, which is consistent with a convex cost structure of child migration.

6 Conclusion

This study provides novel evidence on the effect of family size on child migration, with a particular focus on how son-biasd fertility preferences and migration restrictions jointly shape these dynamics. Our findings demonstrate that gender per se does not seem to affect the child migration rate. Rather, parents with larger families are more likely to leave all children behind. As girls tend to have more siblings due to son-biased fertility preferences, they are less likely to migrate with parents, thereby creating a gender gap in migration rate through family size. Furthermore, we also uncover a gender gap in the family size effect. Specifically, a one-child increase in family size reduces the probability that a girl migrates by 12.9 percentage points, while having little effect on boys. This gender-differential family-size effect thus further widens the gender gap in migration. Critically, we find that this gendered family size effect arises from the interaction between son-biased fertility preferences and a convex migration cost structure induced by migration restrictions. Ironically, while cities with stricter policies often offer superior education, migrant children—especially girls—face greater barriers to accessing these public schools, forgoing significant human capital gains.

Our study thus illuminates how family size shapes the child migration rate in a context of strong son preferences and strict migration policy. Specifically, we show that the gender gap in child migration is driven by two factors: (1) the overrepresentation of girls in larger families created by son-biased fertility preferences, and (2) the convex migration cost structure created by restrictive policies. It has two immediate policy implications. First, because there is no evidence of parental son-biased allocative preferences in child migration, effective intervention should be gender-neutral rather than female-targeted. Second, to mitigate the gender disparity, policy must directly relax the migration restrictions that generate convex costs. In practical terms, this points toward universal subsidies for education and housing. Reducing the non-linear cost barrier would disproportionately benefit larger families, where girls are overrepresented, thereby enabling more migrant children—and especially girls—to migrate with their parents to urban areas and access higher-quality education. In settings where past underinvestment has made the marginal returns to investing in girls particularly high, such reforms could enhance household welfare and support broader economic growth.

References

An, Lei, Yu Qin, Jing Wu, and Wei You. 2024. “The Local Labor Market Effect of Relaxing Internal Migration Restrictions: Evidence from China.” *Journal of Labor Economics* 42 (1): 161–200.

Angrist, Joshua D., Victor Lavy, and Analia Schlosser. 2010. “Multiple Experiments for the Causal Link between the Quantity and Quality of Children.” *Journal of Labor Economics* 28 (4): 773–824.

Bagger, Jesper, Javier A Birchenall, Hani Mansour, and Sergio Urzúa. 2021. “Education, Birth Order and Family Size.” *Economic Journal* 131 (633): 33–69.

Barcellos, Silvia Helena, Leandro S. Carvalho, and Adriana Lleras-Muney. 2014. “Child Gender and Parental Investments In India: Are Boys and Girls Treated Differently?” *American Economic Journal: Applied Economics* 6 (1): 157–189.

Bau, Natalie, Gaurav Khanna, Corinne Low, and Alessandra Voena. 2025. “Traditional Institutions in Modern Times: Dowries as Pensions When Sons Migrate.” *Quarterly Journal of Economics* forthcoming.

Becker, Gary S., and H. Gregg Lewis. 1973. “On the Interaction between the Quantity and Quality of Children.” *Journal of Political Economy* 81 (2): 279–288.

Bhalotra, Sonia, and Damian Clarke. 2019. “Twin Birth and Maternal Condition.” *Review of Economics and Statistics* 101 (5): 853–864.

———, and ———. 2020. “The Twin Instrument: Fertility and Human Capital Investment.” *Journal of the European Economic Association* 18 (6): 3090–3139.

Black, Sandra E, Paul J Devereux, and Kjell G Salvanes. 2005. “The More the Merrier? The Effect of Family Size and Birth Order on Children’s Education.” *Quarterly Journal of Economics* 120 (2): 669–700.

Bratti, Massimiliano, Simona Fiore, and Mariapia Mendola. 2020. “The Impact of Family Size and Sibling Structure on the Great Mexico–USA Migration.” *Journal of Population Economics* 33 (2): 483–529.

Brinch, Christian N., Magne Mogstad, and Matthew Wiswall. 2017. “Beyond LATE with a Discrete Instrument.” *Journal of Political Economy* 125 (4): 985–1039.

Chae, Minhee, Xin Meng, and Sen Xue. 2025. “Fertility, Son Preference, and the Disappearing Gender Gap in Literacy/Numeracy Tests.” *Journal of Population Economics* 38 (Advance Online Publication).

Chen, Shuang. 2020. “Parental Investment After the Birth of a Sibling: The Effect of Family Size in Low-Fertility China.” *Demography* 57 (6): 2085–2111.

Chen, Stacey H., Yen-Chien Chen, and Jin-Tan Liu. 2019. “The Impact of Family Composition on Educational Achievement.” *Journal of Human Resources* 54 (1): 122–170.

Chen, Yuanyuan, and Shuaizhang Feng. 2017. “Quality of Migrant Schools in China: Evidence from a Longitudinal Study in Shanghai.” *Journal of Population Economics* 30 (3): 1007–1034.

——— **, and Wei Fu.** 2023. “Migration Control Policy and Parent–Child Separation Among Migrant Families: Evidence from China.” *Journal of Population Economics* 36 (4): 2347–2388.

Dang, Hai-Anh H., Yang Huang, and Harris Selod. 2020. “Children Left Behind in China: The Role of School Fees.” *IZA Journal of Development and Migration* 11 (1).

——— **, and F. Halsey Rogers.** 2016. “The Decision to Invest in Child Quality over Quantity: Household Size and Household Investment in Education in Vietnam.” *World Bank Economic Review* 30 (1): 104–142.

Davis, Morris A., Jonas D.M. Fisher, and Marcelo Veracierto. 2021. “Migration and Urban Economic Dynamics.” *Journal of Economic Dynamics and Control* 133 104234.

Del Boca, Daniela, Christopher Flinn, and Matthew Wiswall. 2014. “Household Choices and Child Development.” *Review of Economic Studies* 81 (1): 137–185.

Dizon-Ross, Rebecca, and Seema Jayachandran. 2023. “Detecting Mother-Father Differences in Spending on Children: A New Approach Using Willingness-to-Pay Elicitation.” *American Economic Review: Insights* 5 (4): 445–459.

Ebenstein, Avraham. 2010. “The “Missing Girls” of China and the Unintended Consequences of the One Child Policy.” *Journal of Human Resources* 45 (1): 87–115.

Epstein, Gil S. 2008. “Herd and Network Effects in Migration Decision-Making.” *Journal of Ethnic and Migration Studies* 34 (4): 567–583.

Fan, Jingting. 2019. “Internal Geography, Labor Mobility, and the Distributional Impacts of Trade.” *American Economic Journal: Macroeconomics* 11 (3): 252–288.

Fan, Yi, Junjian Yi, Ye Yuan, and Junsen Zhang. 2018. “The Glorified Mothers of Sons: Evidence from Child Sex Composition and Parental Time Allocation in Rural China.” *Journal of Economic Behavior & Organization* 145 249–260.

Gai, Qingen, Naijia Guo, Bingjing Li, Qinghua Shi, and Xiaodong Zhu. 2025. “Rural Pensions, Labor Reallocation, and Aggregate Income: An Empirical and Quantitative Analysis of China.” *Econometrica* 93 (5): 1663–1696.

Gao, Xuwen, Wenquan Liang, Ahmed Mushfiq Mobarak, and Ran Song. 2025. “Daughters Left Behind: How Trade Liberalization Harms Girls in China When Government Restricts Migration.” *Unpublished Manuscript*.

Guo, Naijia, Shuangxin Wang, and Junsen Zhang. 2024. “The Short- and Long-Run Impacts of Free Education on Schooling: Direct Effects and Intra-Household Spillovers.” *Economic Journal* 134 (663): 2876–2911.

Guo, Rufei, Junjian Yi, and Junsen Zhang. 2022. "The Child Quantity–Quality Trade-Off." In *Handbook of Labor, Human Resources and Population Economics*, edited by Klaus F. Zimmermann, 1–23, Cham: Springer International Publishing.

———, ———, ———, and Ning Zhang. 2025. "Rationed Fertility: Treatment Effect Heterogeneity in the Child Quantity-Quality Trade-Off." *Journal of Political Economy* 133 (10): 3349–3386.

Huang, Wei, Xiaoyan Lei, and Ang Sun. 2021. "Fertility Restrictions and Life Cycle Outcomes: Evidence from the One-Child Policy in China." *The Review of Economics and Statistics* 1–17.

Huang, Zibin, Xu Jiang, and Ang Sun. 2024. "Fertility and Delayed Migration: How Son Preference Protects Young Girls Against Mother–Child Separation." *Journal of Development Economics* 166 103191.

———, and Junsen Zhang. 2023. "School Restrictions, Migration, and Peer Effects: A Spatial Equilibrium Analysis of Children's Human Capital in China." *Unpublished Manuscript*.

Jayachandran, Seema, and Ilyana Kuziemko. 2011. "Why Do Mothers Breastfeed Girls Less than Boys? Evidence and Implications for Child Health in India." *Quarterly Journal of Economics* 126 (3): 1485–1538.

Jensen, Robert. 2003. "Equal Treatement, Unequal Outcomes? Generating Sex Inequality Through Fertility Behavior." mimeo, Harvard University, Cambridge, MA.

Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t -Ratio Inference for IV." *American Economic Review* 112 (10): 3260–3290.

Li, Hongbin, Junsen Zhang, and Yi Zhu. 2008. "The Quantity-Quality Trade-Off of Children In a Developing Country: Identification Using Chinese Twins." *Demography* 45 (1): 223–243.

Lin, Carl, Yan Sun, and Chunbing Xing. 2021. "Son Preference and Human Capital Investment among China's Rural-urban Migrant Households." *Journal of Development Studies* 57 (12): 2077–2094.

Meng, Xin. 2012. "Labor Market Outcomes and Reforms in China." *Journal of Economic Perspectives* 26 (4): 75–102.

Mincer, Jacob. 1978. "Family Migration Decisions." *Journal of Political Economy* 86 (5): 749–773.

NBS, UNICEF, and UNFPA. 2023. "What the 2020 Census Can Tell Us About Children in China: Facts and Figures." technical report, National Bureau of Statistics of China, UNICEF China, UNFPA China.

Nevo, Aviv, and Adam M Rosen. 2012. "Identification With Imperfect Instruments." *Review of Economics and Statistics* 94 (3): 659–671.

Oliveira, Jaqueline. 2016. "The Value of Children: Inter-generational Support, Fertility, and Human Capital." *Journal of Development Economics* 120 1–16.

Oster, Emily. 2019. "Unobservable Selection and Coefficient Stability: Theory and Evidence." *Journal of Business & Economic Statistics* 37 (2): 187–204.

Peng, Yinni. 2019. "Getting Rural Migrant Children into School in South China: Migrant Agency and Parenting." *Asian Population Studies* 15 (2): 172–189.

Qian, Nancy. 2008. "Missing Women and the Price of Tea in China: The Effect of Sex-Specific Earnings on Sex Imbalance." *Quarterly Journal of Economics* 123 (3): 1251–1285.

Rosenzweig, Mark R., and Junsen Zhang. 2009. "Do Population Control Policies Induce More Human Capital Investment? Twins, Birth Weight and China's "One-Child" Policy." *Review of Economic Studies* 76 (3): 1149–1174.

Sun, Ang, and Yaohui Zhao. 2016. "Divorce, Abortion, and the Child Sex Ratio: The Impact of Divorce Reform in China." *Journal of Development Economics* 120 53–69.

UNESCO. 2019. *Global Education Monitoring Report 2019: Migration, Displacement and Education – Building Bridges, Not Walls*. 2nd edition, Paris, France: United Nations Educational, Scientifc and Cultural Organization.

Wang, Chunchao, Chenglei Zhang, Jinlan Ni, Haifeng Zhang, and Junsen Zhang. 2019. "Family Migration in China: Do Migrant Children Affect Parental Settlement Intention?" *Journal of Comparative Economics* 47 (2): 416–428.

Wang, Xiaobing, Yu Bai, Linxiu Zhang, and Scott Rozelle. 2017. "Migration, Schooling Choice, and Student Outcomes in China." *Population and Development Review* 43 (4): 625–643.

Wang, Xuebo, and Junsen Zhang. 2018. "Beyond the Quantity–Quality Tradeoff: Population Control Policy and Human Capital Investment." *Journal of Development Economics* 135 222–234.

Wang, Zhiling, and Lu Chen. 2019. "Destination Choices of Chinese Rural–Urban Migrant Workers: Jobs, Amenities, and Local Spillovers." *Journal of Regional Science* 59 (3): 586–609.

Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. MIT Press.

Yum, Minchul. 2023. "Parental Time Investment and Intergenerational Mobility." *International Economic Review* 64 (1): 187–223.

Zhang, Jipeng, and Chong Lu. 2019. "A Quantitative Analysis on the Reform of Household Registration in Chinese Cities." *China Economic Quarterly (in Chinese)* 18 (4): 1509–1530.

Zhang, Junsen. 2017. "The Evolution of China's One-Child Policy and Its Effects on Family Outcomes." *Journal of Economic Perspectives* 31 (1): 141–160.

Zhao, Jianmei, and Hai Zhong. 2019. "A Demographic Factor as a Determinant of Migration: What Is the Effect of Sibship Size on Migration Decisions?" *Journal of Demographic Economics* 85 (4): 321–345.

Tables and Figures

Table 1: Demographic and Socio-Economic Characteristics of Children

	Full sample			No. of children $= 2$			No. of children ≥ 3		
	Full	Boy	Girl	Pooled	Boy	Girl	Pooled	Boy	Girl
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Child characteristics									
Prop. child migration	0.69	0.69	0.69	0.70	0.70	0.70	0.65	0.67	0.63
No. of children	2.11	2.09	2.13	2.00	2.00	2.00	3.09	3.07	3.10
Prop. boy	0.50	1.00	0.00	0.51	1.00	0.00	0.41	1.00	0.00
Prop. primary school age	0.53	0.50	0.57	0.53	0.51	0.55	0.60	0.47	0.69
Child age	5.95	5.70	6.21	5.89	5.71	6.09	6.47	5.59	7.08
Relative birth order	0.50	0.59	0.40	0.50	0.59	0.41	0.49	0.66	0.37
Panel B: Parental characteristics									
Maternal age	31.35	31.33	31.36	31.29	31.29	31.29	31.83	31.76	31.88
Paternal age	33.40	33.33	33.46	33.32	33.27	33.36	34.09	33.99	34.17
Maternal age at second birth	27.24	27.13	27.34	27.45	27.31	27.60	25.37	25.20	25.48
Prop. maternal education is high school or above	0.22	0.22	0.23	0.24	0.23	0.25	0.10	0.09	0.10
Prop. paternal education is high school or above	0.27	0.26	0.27	0.28	0.27	0.29	0.15	0.14	0.15
Panel C: Household characteristics									
Prop. both parents migrate	0.90	0.90	0.90	0.90	0.90	0.89	0.90	0.89	0.90
Prop. grandparent migrate	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Duration of migration in years	5.22	5.22	5.22	5.22	5.23	5.22	5.23	5.17	5.27
Prop. willing to settle	0.61	0.61	0.61	0.61	0.62	0.61	0.57	0.57	0.57
Prop. inter-province migration	0.30	0.30	0.30	0.30	0.30	0.30	0.35	0.36	0.33
Prop. inter-prefecture migration	0.53	0.54	0.53	0.54	0.54	0.53	0.51	0.49	0.52
Observations	38,829	19,276	19,553	34,826	17,631	17,195	4,003	1,645	2,358

Notes: Data is from the 2016 China Migrants Dynamic Survey (CMDS). Means or proportions are reported.

Table 2: First Stage: Effect of Twinning at Second Parity on Family Size

	First-Stage Regressions		
	Full OLS	Boy OLS	Girl OLS
	(1)	(2)	(3)
Twin	0.999***	1.063***	0.959***
	(0.038)	(0.046)	(0.042)
Boy	-0.039***		
	(0.003)		
Sample mean of the outcome variable	2.112	2.091	2.133
R-squared	0.243	0.258	0.238
Observations	38,829	19,276	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS).All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.

***p<.01, **p<.05, *p<.10.

Table 3: The Effect of Family Size on Child Migration

	Full OLS	Full 2SLS	Boy OLS	Girl OLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
No. of children	-0.025***	-0.079**	-0.008	-0.035***	-0.011	-0.129**
	(0.009)	(0.036)	(0.012)	(0.011)	(0.045)	(0.052)
Boy	-0.000	-0.002				
	(0.004)	(0.004)				
Gender Equality Test						
Diff in no. of children			$p = 0.025$		$p = 0.079$	
Kleibergen-Paap F-statistic		706.753			542.337	512.816
Sample mean of the outcome variable	0.693	0.693	0.695	0.691	0.695	0.691
(Centered) R-squared	0.289	0.287	0.285	0.294	0.285	0.290
Observations	38,829	38,829	19,276	19,553	19,276	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

Table 4: Reduced-Form Regressions and Partial Instrument Validity Test

	Reduced-Form Regressions			Partial Instrument Validity Test		
	Full OLS	Boy OLS	Girl OLS	Full OLS	Boy OLS	Girl OLS
	(1)	(2)	(3)	(4)	(5)	(6)
Twin	-0.079**	-0.012	-0.124**	-0.056	-0.004	-0.094*
	(0.035)	(0.048)	(0.049)	(0.037)	(0.049)	(0.050)
No. of children				-0.023**	-0.007	-0.031***
				(0.010)	(0.012)	(0.011)
Boy	0.001			-0.000		
	(0.004)			(0.004)		
Sample mean of the outcome variable	0.693	0.695	0.691	0.693	0.695	0.691
R-squared	0.288	0.285	0.294	0.289	0.285	0.295
Observations	38,829	19,276	19,553	38,829	19,276	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

Table 5: Heterogeneous Family Size Effect on Child Migration by Child Age

	Age 0–5			Age 6–12			Age 13–18		
	Full 2SLS	Boy 2SLS	Girl 2SLS	Full 2SLS	Boy 2SLS	Girl 2SLS	Full 2SLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
No. of children	0.022	0.129	-0.064	-0.103***	-0.048	-0.144***	0.025	0.034	0.023
	(0.083)	(0.103)	(0.119)	(0.039)	(0.052)	(0.055)	(0.043)	(0.070)	(0.054)
Boy	-0.009*			0.004			0.037***		
	(0.006)			(0.006)			(0.009)		
Gender Equality Test									
Diff in no. of children		$p = 0.221$			$p = 0.202$			$p = 0.903$	
Kleibergen-Paap F-statistic	363.368	88.798	336.866	756.104	552.274	393.920	852.281	528.792	410.949
Sample mean of the outcome variable	0.713	0.710	0.716	0.675	0.679	0.671	0.618	0.630	0.607
(Centered) R-squared	0.303	0.291	0.316	0.283	0.285	0.284	0.283	0.279	0.290
Observations	18,063	9,588	8,475	20,766	9,688	11,078	13,097	6,264	6,833

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). The 0-5 and 6-12 age groups are from our main analytical child-level sample, where children are from households without children aged above 12. The 13-18 age group belongs to households without any child older than 18. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.

***p<.01, **p<.05, *p<.10.

Table 6: Heterogeneous Family Size Effects on Child Migration by Maternal Education Attainment

	Maternal Education: Below High School			Maternal Education: High School or Above		
	Full 2SLS	Boy 2SLS	Girl 2SLS	Full 2SLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
No. of children	-0.089**	-0.010	-0.144**	-0.043	-0.023	-0.060
	(0.043)	(0.055)	(0.059)	(0.065)	(0.080)	(0.095)
Boy	-0.003			-0.001		
	(0.005)			(0.008)		
Gender Equality Test						
Diff in no. of children		$p = 0.089$			$p = 0.768$	
Kleibergen-Paap F-statistic	434.017	308.306	348.474	657.051	2,540.408	219.906
Sample mean of the outcome variable	0.687	0.689	0.685	0.711	0.713	0.710
(Centered) R-squared	0.284	0.283	0.285	0.310	0.307	0.321
Observations	30,096	15,061	15,035	8,733	4,215	4,518

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

Table 7: Mechanism: Stringentness of *Hukou* Policies in Destination

	Full 2SLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)
No. of children	-0.023	-0.044	-0.005
	(0.054)	(0.069)	(0.082)
No. of children × Strict	-0.160*	0.019	-0.283**
	(0.085)	(0.115)	(0.118)
Boy	-0.002		
	(0.005)		
Gender Equality Test			
Diff in no. of children		$p = 0.714$	
Diff in no. of children × strict		$p = 0.066$	
Diff in (no. of children + no. of children × strict)		$p = 0.035$	
Kleibergen-Paap F-statistic	182.682	170.926	83.896
Sample mean of the outcome variable	0.682	0.686	0.678
(Centered) R-squared	0.277	0.275	0.280
Observations	25,816	12,788	13,028

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). As the index is only available for 120 prefectural cities, not every city in the CMDS matches with a prefectural index. Hence, only matched observations are kept, resulting in a smaller sample: 25,816 observations, which account for roughly 66.49% of the main analytical sample. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Besides, we also include in the regressions an indicator for strict destination *hukou* policies. Standard errors are clustered at the household level.
***p<.01, **p<.05, *p<.10.

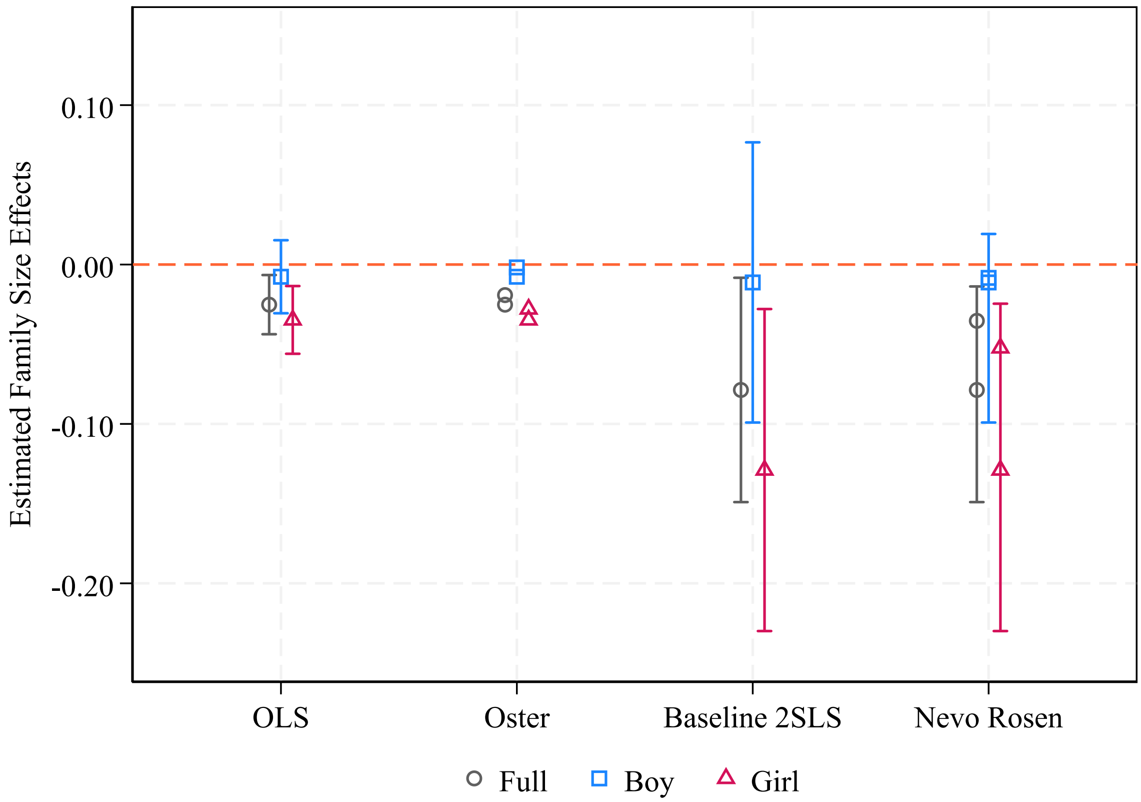


Figure 1: Bounding Family Size Effects

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). The figure plots the estimates of family size effects based on the full sample, the sub-sample of boys, and the sub-sample of girls, respectively. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors are clustered at the household level. 95% confidence intervals are reported.

Online Appendix

Appendix A Models

A.1 Main Model—Son-Biased Fertility Preferences

In this illustrative model, parents value boys' and girls' quality *equally* but the demand for boys drives larger sibship sizes for girls such that it affects investment through the channel of family size.

A.1.1 Setup

Endowments. The household consists of a migrant parent and n children. Suppose that the migrant parent has exogenous income Y. The parent chooses family consumption c and average investment in a child's education quality e.

Child Human Capital Production. To access higher quality education, the child has to migrate to the city. Let child human capital be:

$$h(e) = \theta_0 + \theta_1 e.$$

θ_0 may be interpreted as the human capital of a left-behind child who benefits from free schooling of quality $e = 0$ in the rural hometown. θ_1 captures the returns to investment when a child migrates to the city and thus needs to pay higher schooling fees for higher quality education $e > 0$.

Education Cost. We assume that education cost per unit of quality, $\gamma_{\mathbb{C}}(n)$ in city $\mathbb{C}$, is convex in n: $\gamma'_{\mathbb{C}}(n) > 0$ and $\gamma''_{\mathbb{C}}(n) > 0$. This is based on the fact that it is harder for larger families to enroll all of their children in public schools due to capacity constraints. Thus, some children may go to the more expensive and lower quality migrant or private schools (Chen and Fu, 2023; Wang et al., 2017), which makes it costlier to achieve a given quality of education for all children.[A1,A2]

[A1] From the sub-survey of 2010 CMDS, parents of migrant children in public (migrant or private) schools spent RMB 780.886 (RMB 2,525.659) per child on tuition. The difference is statistically significant at the 1% level.

[A2] Specifically, for a two-child migrant family, one child might enroll in a destination public school through lotteries or network ("*guanxi*") (Kuhn and Shen, 2015; Peng, 2019), but a second child may be forced into an expensive private school due to filled quotas or diminishing benefits of migrants' social network (Epstein, 2008), causing marginal costs to rise steeply. For anecdotal evidence, see https://www.theguardian.com/world/2010/mar/15/china-migrant-workers-children-education and https://www.cq.gov.cn/hdjl/cqhlwdc/lyxd1/202310/t20231027_12485462.html. Moreover, because attending urban schools requires residing in the city, housing costs constitute a significant component of education costs. Migrants are typically live in overcrowded single dwelling units (Wang and Chen, 2019). A family with one child might endure these conditions, but additional children—especially of different genders or older ages—create an urgent need for more space, often forcing a move to a larger and costlier rental. This represents a lump-sum increase

Household Decisions. The parent maximizes utility over consumption and overall human capital of children, where the parent derive a *equal* utility from a boy's or a girl's quality:

$$\ln(c) + \alpha \ln(nh),$$

subject to the budget constraint:

$$c + \gamma_{\mathbb{C}}(n)e = Y.$$

Solving the model, education quality of a child is given by:

$$e = \max\left\{0, \frac{1}{1+\alpha}\left[\frac{\alpha Y}{\gamma_{\mathbb{C}}(n)} - \frac{\theta_0}{\theta_1}\right]\right\}.$$

There is thus a threshold income level—denoted as $Y_T(n, \gamma_{\mathbb{C}}) = Y_T(n, \gamma_{\mathbb{C}}(n))$ for simplicity—above which parents migrate with children and invest positively in their education quality, $e > 0$:

$$Y_T(n, \gamma_{\mathbb{C}}) = \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma_{\mathbb{C}}(n).$$

As Y_T is independent of child gender, Hypothesis 2 is obvious: conditional on the same n, boys and girls have equal likelihood of migrating.[A3] Moreover, the threshold income increases in n, generating Hypothesis 1 (see Figure A2). We next show that the marginal effect of family size on child migration is greater for girls than for boys (Hypothesis 3). Recall that, in this model, the parent has son-biased fertility preferences rather than son-biased allocative preferences.[A4] Thus, the parent treats boys' and girls' quality equally but makes son-biased fertility choices. Building on Basu and De Jong (2010) and Jensen (2003), we start by defining such choices of childbearing:

Definition A1 (Son-biased fertility choice.) *Couples continue childbearing until they attain a desired target number of boys, or reach the maximum number of children.*

As a result, couples would have shorter birth intervals, and are less likely to use contraception, and eventually have more children (girls) (Barcellos et al., 2014). We therefore get the following lemma:

Lemma A1 (Girls end up with more siblings.) *The expected number of siblings for female children is greater than expected number of siblings for male children.*

See Basu and De Jong (2010) and Jensen (2003) for a formal proof of Lemma A1. Next, define family sizes $n^{\mathcal{S}}$ for small families and $n^{\mathcal{L}}$ for large families: $n^{\mathcal{S}} < n^{\mathcal{L}}$, and note that the threshold

in fixed costs far beyond a linear rate, particularly in mega cities.

[A3] Figure A2a provides a graphical representation. For a given family size, the level of left behind is always Θ.

[A4] In spirit of Jensen (2003), instead of modeling the underlying determinants of the son-biased fertility preference, we take such preference as given and demonstrate its key implications.

income increases at an accelerating rate due to convexity: $\frac{\partial^2 Y_T(n,\gamma_{\mathbb{C}})}{\partial n^2} = \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma''_{\mathbb{C}}(n) > 0$. It follows that Y_T increases by a greater amount for larger families than for smaller families when n increase by one unit (see Figure A2b):

$$\left.\frac{\partial Y_T(n,\gamma_{\mathbb{C}})}{\partial n}\right|_{n^{\mathcal{S}}} < \left.\frac{\partial Y_T(n,\gamma_{\mathbb{C}})}{\partial n}\right|_{n^{\mathcal{L}}} \Rightarrow \left.\frac{\partial Y_T(n,\gamma_{\mathbb{C}})}{\partial n}\right|_{\text{boys}} < \left.\frac{\partial Y_T(n,\gamma_{\mathbb{C}})}{\partial n}\right|_{\text{girls}}. \tag{A1}$$

Thus, given Lemma A1 which implies that girls are disproportionately represented in larger families, we would observe gender differences in the family size effect—boys would be less likely to be left-behind compared to girls when n increases. As a result, the gender gap in child migration rate is widened. We thus deliver Hypothesis 3.[A5]

It is also straightforward to see that, for a given n, the threshold income is higher in cities with more stringent migration restrictions compared to cities with more relaxed migration restrictions:

$$Y_T(n,\gamma_{\mathbb{S}}) = \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma_{\mathbb{S}}(n) > \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma_{\mathbb{R}}(n) = Y_T(n,\gamma_{\mathbb{R}}).$$

Thus, the probability that children are left behind when parents migrate to a stringent city is greater than when parents migrate to a relaxed city, generating Hypothesis A1 (see Figure A3a).

Hypothesis A1 (Migration restrictions.) *Children whose parents migrated to more strict cities have a lower likelihood of migrating with parents.*

Finally, the rate at which the threshold income increases in n is greater for more stringent cities: $\frac{\partial Y_T(n,\gamma_{\mathbb{S}})}{\partial n} = \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma'_{\mathbb{S}}(n) > \frac{1}{\alpha}\frac{\theta_0}{\theta_1}\gamma'_{\mathbb{R}}(n) = \frac{\partial Y_T(n,\gamma_{\mathbb{R}})}{\partial n}$. This implies that the probability that a child migrates will decrease to a greater extent when there is an increase in n, when the parent migrates to a strict city compared to a relaxed city. Moreover, as shown above, the rate at which the threshold income increases in n is greater for larger than for smaller family sizes. As girls tend to have more siblings, it follows that the gendered family size effects are exacerbated in the presence of more stringent migration restrictions, giving rise to Hypothesis A2 (see Figure A3b).

Hypothesis A2 (Family size and migration restrictions.) *The gendered family size effects are exacerbated in the presence of more stringent migration restrictions.*

[A5] Figure A2b provides a graphical representation. When girls end up in larger families, the level of left-behind is higher for girls than for boys, i.e., $(\Theta^{\mathcal{L}} - \Theta^{\mathcal{S}}) > 0$. Moreover, the diseconomies of scale in migration costs—induced by migration restrictions—make the marginal increase in the level of left-behind higher for girls than for boys following a one-unit increase in family size, i.e., $\Delta^{\mathcal{L}} > \Delta^{\mathcal{S}}$. Consequently, the initial gender gap in left-behind rate is widened, i.e., $\left(\Theta^{\mathcal{L}'} - \Theta^{\mathcal{S}'}\right) > (\Theta^{\mathcal{L}} - \Theta^{\mathcal{S}})$.

A.2 Alternative Model—Son-Biased Allocative Preferences

We now modify the model to allow the parent to have different valuations for sons' and daughters' human capital, where parents derive a higher utility from a boy's quality. Let e_g and $h_g = \theta_0 + \theta_1 e_g$, respectively, denote investment in education quality and human capital of a child of gender $g = S, D$. Assume further that the proportion of sons in the family is $q = 0.5$ as we observe in our data (see Table 1). Son-biased allocative preference is captured by the fact that the parent puts a greater weight on sons than on daughters: $\alpha > 1 - \alpha$.

The parent maximizes utility over consumption and human capital of sons and daughters:

$$\ln(c) + \alpha \ln(qnh_S) + (1 - \alpha) \ln((1 - q)nh_D),$$

subject to the budget constraint:

$$c + \gamma_{\mathbb{C}}(n)qe_S + \gamma_{\mathbb{C}}(n)(1 - q)e_D = Y.$$

Solving the model, the human capital of sons and daughters are, respectively, given by:

$$h_S = \frac{\alpha}{\gamma_{\mathbb{C}}(n)}[\theta_1 Y + \gamma_{\mathbb{C}}(n)\theta_0],$$

$$h_D = \frac{1 - \alpha}{\gamma_{\mathbb{C}}(n)}[\theta_1 Y + \gamma_{\mathbb{C}}(n)\theta_0].$$

It follows that there are threshold income levels above which parents migrate with sons ($e_S > 0$ or $h_S > \theta_0$) and daughters ($e_D > 0$ or $h_D > \theta_0$), respectively:

$$Y_S = \frac{(1 - \alpha)\theta_0}{\alpha\theta_1}\gamma_{\mathbb{C}}(n),$$

$$Y_D = \frac{\alpha\theta_0}{(1 - \alpha)\theta_1}\gamma_{\mathbb{C}}(n).$$

Whereas Hypotheses 1, 3, and A2 carry over in this model, Hypothesis 2 does not carry over. Specifically, since $1 - \alpha < \alpha$, we have $Y_S < Y_D$ for a given n. Thus, son preference on child quality implies son-biased resource allocation, and that parents will be more likely to migrate with sons than with daughters, even if n were the same. We do not find support for this in our analytic sample (see Section 4).

A.3 Policy Implications

The two models generate different policy implications. In the first model of son-biased fertility preference, gender-neutral education subsidies may help mitigate the gendered family size effects. By reducing $\gamma_{\mathbb{C}}(n)$, the slope of $Y_T(n, \gamma_{\mathbb{C}})$ decreases such that the gap in the threshold incomes between smaller and larger families also decreases (see Figure A3b). This implies that whereas the relaxation of migration restrictions in the form of gender-neutral education subsidies encourages both small and larger families to migrate with children, such incentives are stronger for larger than for smaller families. As girls tend to belong to larger families, they experience a much larger decrease in the probability that they would be left-behind when migration restrictions are relaxed.

In the second model of son-biased allocative preference, the family size effects arise due to the direct effect of allocative preference on child quality. Thus, this model predicts that even within the same family, parents may migrate with sons but not daughters. In such case, gender-targeted education subsidies that reduce $\gamma_{\mathbb{C}}(n)$ only for daughters may help mitigate the gendered family size effects. Should we observe families with mixed-sex children are more likely to bring their sons but not daughters to cities *ceteris paribus*, the gendered policy implication would hold. However, our family fixed-effect estimates show that the within-household gender effect is statistically indistinguishable from zero (see Section 4.4). Thus, the gender neutral policy from the first model seems more pertinent in our context.

Appendix B Robustness Checks

We now discuss several sensitivity analyses on our main gendered family size effects including (i) bounding family size effects: accounting for selection on unobservables using selection on observables and relaxing the exclusion restriction, (ii) additional specifications: two-step estimation and control function approach, and (iii) extrapolating from LATE to estimate ATE.

B.1 Bounding Family Size Effects

Accounting for Selection on Unobservables Using Selection on Observables. We follow Oster (2019) to bound the family size effects using selection on observed variables as a guide to selection on unobserved variables in the OLS model (1). This method builds on the observation that omitted variable bias is often deemed to be limited if a coefficient is stable when observed controls are added in a regression. To derive the bias-adjusted treatment effect, we adopt two recommended assumptions from Oster (2019). The first assumption is that there is equal selection on observables and unobservables, that is, the observables and unobservables are equally related to the treatment, which is sibship size in our case. The second assumption is that the attainable R^2 in a regression with a full set of (observable and unobservable) controls, R_{max}, is equal to 1.3 times the estimated R^2 from a regression with observable controls only, $\hat{R}_{OLS}$, as per specification (1). Under these two assumptions, the identified set lies between the OLS estimate from (1), $\hat{\beta}_1^{OLS}$, and the bias-adjusted estimate, $\beta_1^* = \hat{\beta}_1^{OLS} - \left[\hat{\beta}_1^0 - \hat{\beta}_1^{OLS}\right] \frac{R_{max} - \hat{R}_{OLS}}{\hat{R}_{OLS} - \hat{R}_0}$, where $\hat{\beta}_1^0$ and $\hat{R}_0$ are, respectively, the estimated treatment effect and R^2 from a regression of child migration on sibship size without any controls. Figure 1 plots OLS estimates as well as the bounds à la Oster (2019). From the figure, the identified sets of β_1 are fairly tight and are consistent with our inference that while the probability that a girl migrates is negatively associated with sibship size, the association between the probability that a boy migrates and family size is very small and close to zero.

Relaxing the Exclusion Restriction. For the $Twin_{jp}$ instrument to be valid in the 2SLS model (2)–(3), it has to satisfy the exclusion restriction: $Twin_{jp} \perp \varepsilon_{ijp}$. However, the exclusion restriction may be violated because twins have lower average birth weight than singletons (Rosenzweig and Zhang, 2009). Parents may thus reallocate resources from twins to singletons, which would dampen the (negative) associations between family size and parental investments in children. We thus relax the assumption that $Twin_{jp} \perp \varepsilon_{ijp}$ and estimate bounds on family size effects to provide more compelling evidence of family size effects on child migration (Bhalotra and Clarke, 2020). In particular, we follow the method proposed by Nevo and Rosen (2012), which allows for the IV to be correlated with the error term in the second stage regression. Under the relatively weak assumptions that (i) the correlation between the instrument and ε_{ijp} has the same sign as the correlation between the endogenous regressor and ε_{ijp} and (ii) the instrument is less endogenous than the endogenous regressor, we can derive informative bounds for the causal effects of family size on child migration based on set identification.

In our context, we argue that FS_{jp} and ε_{ijp} could be positively correlated because parents with a strong preference for old age support from children could potentially choose to have more children and also be more likely to migrate with children. This is consistent with the seemingly upward bias from OLS estimates compared to 2SLS estimates in Table 3. Conversely, we believe that $Twin_{jp}$ and ε_{ijp} could be negatively correlated because the presence of $Twin_{jp}$ in the family imposes simultaneous and greater demands on parental time, which may result in a lower ability to cater for and thus migrate with other children. To satisfy the first assumption (i), we specify our instrument as $-Twin_{jp}$. In that case, there is also a negative correlation between FS_{jp} and $-Twin_{jp}$ as can be inferred from our first stage regression coefficients reported in Table 3. Nevo and Rosen (2012) show that in such situation, the family size effect would be bounded by the 2SLS and OLS estimates: $\hat{\beta}_1^{2SLS} \leq \beta_1 \leq \hat{\beta}_1^{OLS}$. If the second assumption (ii) additionally holds, then we can obtain tighter bounds: $\hat{\beta}_1^{2SLS} \leq \beta_1 \leq \hat{\beta}_1^{NR}$, where $\hat{\beta}_1^{NR}$ is a 2SLS estimator that employs $NR = \sigma_{FS} Twin_{jp} - \sigma_{Twin} FS_{jp}$ as an IV, and σ_{var} denotes the standard deviation of

$var = Twin_{jp}, FS_{jp}$.

Figure 1 reports the point estimates and the 95% confidence intervals from OLS and 2SLS estimates. From the figure, the presence of negative family size effects on child migration are confirmed for girls but not for boys. Note that Nevo and Rosen (2012) argue that the effects are always bounded by $\hat{\beta}_1^{OLS}$, $\hat{\beta}_1^{2SLS}$ or $\hat{\beta}_1^{NR}$. Given that all estimates including the 95% confidence intervals are always negative for girls, the above conjectures on the direction of the various correlations effectively do not matter for the purpose of confirming the presence of negative family size effects for girls. Our results and inference that family size effects matter for girls but not for boys are thus robust to the relaxation of the exclusion restriction.

B.2 Alternative Specifications

Two-Step Estimation. It is well-known that birth order configuration tends to be jointly determined with family size. We thus perform an additional robustness check to identify birth order effects separately from family size effects using a two-step estimation approach (Bagger et al., 2021; Bratti et al., 2020). In the first step, we use family fixed effect (FFE) models—that exploit within-family variation—to identify birth order effects. Note that family size and other household level predictors (including unobserved preferences) would be differenced out in the FFE models. We then predict child migration net of birth order effects. In the second step, we apply 2SLS as described above but using the predicted child migration net of birth order effects as outcome variable. As the outcome variable is now predicted from a regression, we block-bootstrap the standard errors, clustered at the household level. From Appendix Table A7, the results and inferences from the two-step estimation are similar to those from the main specification in Table 3.

Control Function Approach. We now take the discrete nature of migration into account and estimate marginal effects using Probit models and a control function (CF) approach (Wooldridge, 2010). From Columns (7)–(9) of Appendix Table A7, the marginal effects are very similar to those from the linear models, which boosts confidence in the robustness of the 2SLS estimates.

B.3 Moving Beyond LATE and Estimating ATE

We follow Brinch et al. (2017) to estimate MTEs and extrapolate to obtain ATEs. As the MTE framework builds on binary treatment, we start by replicating our 2SLS model (2)–(3) by replacing FS with D^{FS}, an indicator that takes unity if the family has three or more children and 0 if the family has two children, and perform subsample analyses for boys and girls separately. The inferences using D^{FS} as treatment (Table A8) are similar to those that use FS (Table 3).

Next, we reframe the econometric model as a generalized Roy model:

$$M_{ijp} = (1 - D^{FS}_{jp})M^0_{ijp} + D^{FS}_{jp}M^1_{ijp}, \tag{A2}$$
$$M^t_{ijp} = \mathbf{X}'\boldsymbol{\iota}^t + U^t_{ijp}, \tag{A3}$$
$$D^{FS}_{jp} = \mathbb{I}\left\{v(\mathbf{X}, \mathbf{Z}_{jp}) > V^D_{ijp}\right\}, \tag{A4}$$

where $\mathbf{X}$ is augmented to include all covariates, $\mathbf{Z}_{jp}$ are the IVs, and $t \in \{0, 1\}$. The conditional expectations of M^t_{ijp} for $t = 0, 1$ are defined:

$$\mathbb{E}(M^t_{ijp} \mid \mathbf{X} = \mathbf{x}, V^D_{ijp} = c) = \mathbf{x}'\boldsymbol{\iota}^t + \mathbb{E}(U^t_{ijp} \mid V^D_{ijp} = c) \equiv \mathbf{x}'\boldsymbol{\iota}^t + k_t(c).$$

The MTE can then be expressed as a function of $\mathbf{x}$ and $k_t(c)$:

$$\Delta^{MTE}(\mathbf{x}, c) = \mathbb{E}(M^1_{ijp} \mid \mathbf{X} = \mathbf{x}, V^D_{ijp} = c) - \mathbb{E}(M^0_{ijp} \mid \mathbf{X} = \mathbf{x}, V^D_{ijp} = c) = \mathbf{x}'(\boldsymbol{\iota}^1 - \boldsymbol{\iota}^0) + k_1(c) - k_0(c).$$

The MTE can be interpreted as the expected treatment effects for individuals who are indifferent between having at least three children and having two, with $\mathbf{X} = \mathbf{x}$ and $\mathbb{P}(D^{FS}_{ijp} = 1 \mid \mathbf{X}, \mathbf{Z}_{jp}) = c$. Following the literature, we assume that $k_t(c) = \alpha_t\left(c - \frac{1}{2}\right)$, where $k_t(c)$ is a linear function of c and α_t, and may be interpreted as the coefficient of an "inverse Mills ratio" type expression in the canonical Heckit model. As it is challenging to identify MTE with $Twin_{jp}$ as the only IV (there are no never-takers as families with twins at second birth have at least three kids), we use an indicator for having the same gender for the first two births as an additional IV to $Twin_{jp}$ (see Table A8).

Figure A4 plots the estimated MTEs for boys and girls. In our context, the MTEs capture the average effects of being born in a household with three or more children for girls or boys on the margin between migrating and being left-behind. These margins correspond to percentiles of the distribution of the unobserved resistance, V^D_{ijp}. Following Andresen (2018) and Mogstad and Torgovitsky (2018), we next express the ATEs as a weighted average of MTEs, assigning the same weight on the MTE at each value of V^D_{ijp}. The ATE for boys is negligible (-0.006, p >0.1) while the ATE for girls is -0.111 ($p < 0.05$), which is consistent with gendered family size effects.

Appendix C A Non-Linear Specification of Family Size Effect

This appendix outlines the econometric approach to address endogeneity in non-linear family size specifications. For the ease of exposition, we replicate equation (5), which outlines the non-linear specification of family size on child migration M_{ijp} à la Bagger et al. (2021):

$$M_{ijp} = \beta_0 + \beta_1 Boy_{ijp} + \phi_3\mathbb{I}\left\{FS_{jp} = 3\right\} + \phi_{4^+}\mathbb{I}\left\{FS_{jp} \geq 4\right\} + \mathbf{X}'_{ijp}\boldsymbol{\Gamma} + \mu_p + \varepsilon_{ijp}, \tag{A5}$$

where FS_{jp} denotes the family size for family j in the destination province p, $\mathbf{X}_{ijp}$ is a vector of control variables, and μ_p represents destination province fixed effects. To address endogeneity issue, we use twinning at second parity and twinning at third or higher parity as the instruments for the indicator variables $\mathbb{I}\left\{FS_{jp}=3\right\}$ and $\mathbb{I}\left\{FS_{jp}\geq 4\right\}$, respectively. It is noteworthy that our sample is restricted to families with at least two children, with a maximum family size of five. Due to the small proportion of households with five children, we combine them with households of four children. Consequently, our analysis employs only two family size dummies, using households with exactly two children as the reference category to prevent collinearity.

In addition, because the instrument no longer just twinning at second parity in the non-linear model, but also additionally includes twin births in the third or higher parity, it is undefined or missing for households with exactly two children. Specifically, twin birth indicators are *missing* for families that are truncated before the relevant parity. In our context, the indicator for twin birth at parity three or higher ($Twin_{jp}^{3+}$) is only observed for families with $FS_{jp}\geq 3$. Thus, for families with $FS_{jp}=2$, this indicator is missing, creating a correlation between the instrument and family size through the truncation mechanism.

In order to address the partially-missing issue for twinning at three or higher parity, we follow Angrist et al. (2010) to construct full-sample instruments by "filling-in" values:

$$\widetilde{Twin}_{jp}^{3^+}=\left(Twin_{jp}^{3^+}-\mathbb{E}\left[Twin_{jp}^{3^+}\mid \mathbf{X}_{ijp},FS_{jp}\geq 3\right]\right)\times\mathbb{I}\left(FS_{jp}\geq 3\right), \tag{A6}$$

where $Twin_{jp}^{3^+}$ is the raw indicator for twin birth at three or higher parity, and $\mathbb{E}\left[Twin_{jp}^{3^+}\mid \mathbf{X}_{ijp},FS_{jp}\geq 3\right]$ is the expected value of twin births conditional on covariates and family size being at least 3.

Appendix D Additional Tables and Figures

Table A1: Educational Performance & Health Status: Migrant *versus* Left-Behind Children

	Full Sample		Girl		Boy	
	Left-Behind	Migrant	Left-Behind	Migrant	Left-Behind	Migrant
Panel A: Selected test score and health outcomes						
Chinese test score (standardized)	69.45	71.07*	72.49	74.00*	66.93	68.41*
	(10.07)	(9.34)	(8.62)	(8.15)	(10.32)	(9.53)
Math test score (standardized)	69.39	70.78*	69.81	71.21*	69.12	70.49 *
	(10.27)	(9.50)	(9.94)	(9.06)	(10.47)	(9.82)
English test score (standardized)	69.30	70.41*	72.55	73.36*	66.60	67.80*
	(10.25)	(9.55)	(8.68)	(8.35)	(10.60)	(9.75)
BMI	18.68	18.96*	18.54	18.68	18.81	19.22*
	(3.18)	(3.42)	(2.80)	(2.90)	(3.49)	(3.84)
Ever hospitalized last year	0.09	0.07*	0.09	0.06*	0.09	0.08
	(0.29)	(0.25)	(0.29)	(0.25)	(0.29)	(0.27)
Overall health status (1:worst; 5:best)	3.93	4.12*	3.87	4.09*	3.98	4.16*
	(0.91)	(0.90)	(0.91)	(0.88)	(0.91)	(0.91)
Shortsighted	0.54	0.62*	0.60	0.68*	0.49	0.57*
	(0.50)	(0.48)	(0.49)	(0.47)	(0.50)	(0.49)
Grit index	9.50	9.49	9.94	9.86	9.14	9.18
	(2.65)	(2.72)	(2.27)	(2.54)	(2.87)	(2.80)
Noncognitive skill	11.77	11.95*	11.84	11.99	11.75	11.94
	(3.32)	(3.40)	(2.93)	(3.19)	(3.59)	(3.53)
Depression index	10.67	10.18*	10.95	10.18*	10.40	10.21
	(4.33)	(4.45)	(4.17)	(4.04)	(4.42)	(4.76)
Panel B: Demographics						
Age	13.61	13.43*	13.57	13.39*	13.64	13.47*
	(1.28)	(1.23)	(1.27)	(1.23)	(1.29)	(1.23)
Single child in the family	0.34	0.33	0.32	0.28*	0.37	0.39
	(0.47)	(0.47)	(0.47)	(0.45)	(0.48)	(0.49)
Boy	0.53	0.52				
	(0.50)	(0.50)				
Mother's education attainment	3.45	3.50	3.48	3.53	3.43	3.49
	(1.83)	(1.80)	(1.86)	(1.78)	(1.82)	(1.82)
Father's education attainment	3.78	4.03*	3.84	4.07*	3.75	3.99*
	(1.81)	(1.89)	(1.85)	(1.90)	(1.78)	(1.89)
Family income status	2.84	3.07*	2.83	3.07*	2.86	3.06*
	(0.60)	(0.47)	(0.58)	(0.46)	(0.62)	(0.49)
Observations	4,400	2,888	2,020	1,360	2,301	1,490

Notes: Data from the China Education Panel Study (2013-2014). Unweighted means and standard deviations (in parentheses) are presented. Mother's and father's education attainment: 1: no schooling; 2: elementary; 3: junior high; 4: technical school; 5: vocational high school; 6: senior high; 7: junior college; 8: bachelor degree; 9: above bachelor degree. Family income status: 1: very poor; 2: somewhat poor; 3: moderate; 4: somewhat rich; 5:very rich. A handful of observations were missing from some of the variables including child gender. Migrant children are defined as those without local *hukou*; Left-behind children defined as those living without at least one of their parents. * indicates that the difference between left-behind and migration children is statistically significantly different at the 5% level.

Table A2: Balance Check: Do Pre-Determined Characteristics Predict the Instrument?

	Dummy for twinning at second parity					
	(1)	(2)	(3)	(4)	(5)	(6)
Maternal age at second birth	0.00003	-0.00024	-0.00021	-0.00022	-0.00024	-0.00023
	(0.00008)	(0.00017)	(0.00017)	(0.00017)	(0.00017)	(0.00017)
Maternal age		0.00116*	0.00034	0.00037	0.00037	0.00038
		(0.00062)	(0.00085)	(0.00086)	(0.00086)	(0.00087)
Maternal age square		-0.00001	-0.00000	-0.00000	-0.00000	-0.00000
		(0.00001)	(0.00001)	(0.00001)	(0.00001)	(0.00001)
Paternal age			0.00126	0.00128	0.00123	0.00130
			(0.00083)	(0.00083)	(0.00083)	(0.00082)
Paternal age square			-0.00002	-0.00002	-0.00002	-0.00002
			(0.00001)	(0.00001)	(0.00001)	(0.00001)
Dummy for maternal education is middle school				-0.00196	-0.00233	-0.00230
				(0.00124)	(0.00147)	(0.00147)
Dummy for maternal education is high school				-0.00029	-0.00121	-0.00119
				(0.00144)	(0.00179)	(0.00179)
Dummy for maternal education is college and above				-0.00045	-0.00180	-0.00182
				(0.00206)	(0.00216)	(0.00216)
Dummy for paternal education is middle school					0.00069	0.00065
					(0.00158)	(0.00158)
Dummy for paternal education is high school					0.00160	0.00161
					(0.00184)	(0.00183)
Dummy for paternal education is college and above					0.00215	0.00219
					(0.00222)	(0.00222)
Dummy for both parents migrant						0.00064
						(0.00109)
Dummy for grandparent migrant						0.00210
						(0.00166)
Dummy for inter-province migration						-0.00133
						(0.00105)
Dummy for inter-prefecture migration						0.00013
						(0.00103)
Duration of migration in years						0.00006
						(0.00009)
Dummy for willing to settle						-0.00004
						(0.00070)
Sample mean of the outcome variable†	0.00412	0.00412	0.00412	0.00412	0.00412	0.00412
R-squared	0.00000	0.00022	0.00030	0.00047	0.00052	0.00071
Observations	38,829	38,829	38,829	38,829	38,829	38,829

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). Standard errors clustered at the household level are in the parentheses.

†The sample mean is the proportion of singleton children who had second-born twin siblings.

****p*<.01, ***p*<.05, **p*<.10.

Table A3: Balance Check by Twinning at Second Parity at Household Level

	Twin Families		Non-Twin Families		Tests of Differences	
	Mean/Prop.	S.D.	Mean/Prop.	S.D.	(1)−(3)	*p*-value
	(1)	(2)	(3)	(4)	(5)	(6)
Maternal age at second birth	27.50	3.83	27.30	4.02	0.21	0.53
Maternal age	31.97	3.98	31.33	4.21	0.65	0.06
Paternal age	34.21	4.11	33.37	4.55	0.84	0.02
Dummy for maternal education is middle school	0.55	0.50	0.63	0.48	-0.09	0.03
Dummy for maternal education is high school	0.22	0.42	0.18	0.39	0.04	0.23
Dummy for maternal education is college and above	0.05	0.22	0.04	0.21	0.01	0.66
Dummy for paternal education is middle school	0.58	0.50	0.63	0.48	-0.05	0.16
Dummy for paternal education is high school	0.25	0.43	0.21	0.41	0.04	0.24
Dummy for paternal education is college and above	0.08	0.27	0.06	0.24	0.02	0.32
Dummy for both parents migrant	0.92	0.28	0.90	0.30	0.02	0.50
Dummy for grandparent migrant	0.08	0.28	0.06	0.23	0.03	0.15
Dummy for inter-province migration	0.25	0.43	0.30	0.46	-0.05	0.16
Dummy for inter-prefecture migration	0.58	0.49	0.54	0.50	0.05	0.25
Duration of migration in years	5.82	4.96	5.22	4.38	0.60	0.09
Dummy for willing to settle	0.63	0.48	0.61	0.49	0.02	0.57

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). The analyses are conducted at household level. We refer to households with and without second-born twins to as twin and non-twin families, respectively. No. of twin families = 153, and no. of non-twin families = 18,711.

Table A4: Family Size Trade-Offs on Households with Only Child and Two or More Children

	Full OLS	Full 2SLS	Boy OLS	Girl OLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
No. of children	-0.027***	-0.036	-0.027***	-0.027***	0.059	-0.168**
	(0.004)	(0.057)	(0.005)	(0.005)	(0.064)	(0.086)
Boy	0.001	-0.001				
	(0.003)	(0.010)				
Gender Equality Test						
Diff in no. of children			$p = 0.995$		$p = 0.021$	
Kleibergen-Paap F-statistic		238.470			291.179	117.066
Sample mean of the outcome variable	0.704	0.704	0.705	0.703	0.705	0.703
(Centered) R-squared	0.312	0.312	0.310	0.315	0.305	0.293
Observations	75,595	75,595	40,788	34,807	40,788	34,807

Notes: Data from the 2016 China Migrants Dynamic Survey (CMDS). We include all only-child households in the data. We then use the occurrence of first-born twins as the instrument. And we assume the relative birth order for the child in the only-child households as 0, i.e., they are regarded as firstborns. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at *first* birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.

***p<.01, **p<.05, *p<.10.

Table A5: Family Size Trade-Offs on Families Completed Fertility Phase

	Youngest Child Aged over Six			Firstborn Child Aged over Six		
	Full 2SLS	Boy 2SLS	Girl 2SLS	Full 2SLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
No. of children	-0.093**	-0.028	-0.139***	-0.085**	-0.025	-0.133**
	(0.036)	(0.047)	(0.051)	(0.041)	(0.052)	(0.060)
Boy	0.001			0.000		
	(0.009)			(0.005)		
Gender Equality Test						
Diff in no. of children	$p = 0.104$			$p = 0.166$		
Kleibergen-Paap F-statistic	982.815	443.680	731.993	443.501	394.475	285.626
Sample mean of the outcome variable	0.669	0.675	0.662	0.690	0.693	0.686
(Centered) R-squared	0.282	0.293	0.276	0.279	0.281	0.279
Observations	6,819	3,731	3,088	27,382	13,819	13,563

Notes: Data from the 2016 China Migrants Dynamic Survey (CMDS). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Note that in the sub-sample of children from households with the youngest child aged over six, all children are in their primary school age. Hence, there is no variation for the schooling age dummy, and we omit the dummy for primary school age from the regressions in columns (1)–(3). Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

Table A6: Sample Restrictions

	Firstborn Children			Children Born Elsewhere		
	Full 2SLS	Boy 2SLS	Girl 2SLS	Full 2SLS	Boy 2SLS	Girl 2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
No. of children	-0.095***	-0.035	-0.136***	-0.088**	0.017	-0.165***
	(0.035)	(0.048)	(0.049)	(0.042)	(0.052)	(0.058)
Boy	-0.006			0.003		
	(0.006)			(0.005)		
Gender Equality Test						
Diff in no. of children		$p = 0.141$			$p = 0.017$	
Kleibergen-Paap F-statistic	1,889.853	793.834	1,124.643	478.986	391.949	372.610
Sample mean of the outcome variable	0.680	0.676	0.683	0.596	0.595	0.596
(Centered) R-squared	0.291	0.297	0.290	0.278	0.277	0.280
Observations	18,804	7,646	11,158	26,570	12,923	13,647

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

Table A7: Alternative Specifications: Two-Step Estimation and Control Function Approach

	First Step Estimation			Second Step Estimation			Control Function Approach		
	Full FFE	Boy FFE	Girl FFE	Full 2SLS	Boy 2SLS	Girl 2SLS	Full CF	Boy CF	Girl CF
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
No. of children				-0.075**	-0.011	-0.125**	-0.073**	-0.015	-0.110***
				(0.030)	(0.045)	(0.049)	(0.032)	(0.047)	(0.042)
Boy	0.000			-0.005			-0.002		
	(0.003)			(0.004)			(0.004)		
Relative birth order	-0.006	0.004	-0.028***				-0.030***	-0.001	-0.055***
	(0.005)	(0.010)	(0.011)				(0.010)	(0.016)	(0.014)
Gender Equality Test									
Diff in no. of children					$p = 0.067$			$p = 0.117$	
Kleibergen-Paap F-statistic				788.129	617.828	596.544			
Sample mean of the outcome variable	0.693	0.695	0.691	0.696	0.692	0.702	0.693	0.695	0.691
(Centered) R-squared	0.011	0.008	0.013	0.288	0.285	0.292	0.259	0.256	0.265
Observations	38,829	19,276	19,553	38,829	19,276	19,553	38,829	19,276	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). Two-step estimation comprise of family fixed effects (FFE) models in the first step – Columns (1), (2), and (3) – and 2SLS in the second step – Columns (4), (5), and (6). We highlight the relative birth order coefficients here, as opposed to in other tables, because they reflect the core of the two-step approach—estimating the impact of family size netting out the effect of birth order through first-stage adjustment. Marginal effects are reported for the control function (CF) approach, which is implemented via a two-step IV-Probit estimation procedure – Columns (7), (8), and (9). The standard errors in Columns (1)–(3) and (7)–(9) are clustered at the household level while standard errors in Columns (4)–(6) are block-bootstrapped at the family level with 200 repetitions. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. However, depending on the specification, all family-level controls are omitted in columns (1)–(3), and relative birth order is omitted from columns (4)–(6).
***p<.01, **p<.05, *p<.10.

Table A8: Family Size Effects with Binary Treatment

	Boy			Girl		
	Both IV (1)	Twin IV Only (2)	Same-sex IV Only (3)	Both IV (4)	Twin IV Only (5)	Same-sex IV Only (6)
Second-stage coefficients:						
No. of children ≥ 3	-0.038	-0.013	-0.179	-0.109***	-0.144**	-0.087**
	(0.049)	(0.052)	(0.148)	(0.034)	(0.056)	(0.043)
First-stage coefficients:						
Twins	0.918***	0.923***		0.845***	0.861***	
	(0.027)	(0.027)		(0.022)	(0.020)	
Same sex	0.045***		0.046***	0.153***		0.155***
	(0.005)		(0.005)	(0.006)		(0.006)
Kleibergen-Paap F-statistic	615.689	1159.208	97.903	1169.191	1874.624	628.230
Hansen J statistics p-value	0.287			0.424		
Centered R-squared	0.285	0.285	0.277	0.293	0.291	0.294
Observations	19,276	19,276	19,276	19,553	19,553	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). All the controls listed in Section 4 are included in the models above. We conduct subsample regressions for boys and girls with different instrument(s) and a binary family size measure. In Columns (1) & (4), both twinning at second parity and having first two children with the same sex serve as instruments, whereas twinning at second parity serves as the only instrument in the Columns (2) & (5) and having first two children with the same sex as the only instrument in the Columns (4) & (6). All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, relative birth order, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. The standard errors are clustered at the household level.
$^{***}p<.01$, $^{**}p<.05$, $^{*}p<.10$.

Table A9: Non-Linear Family Size Effect on Child Migration

	Full 2SLS (1)	Boy 2SLS (2)	Girl 2SLS (3)
No. of children = 3	-0.086**	-0.016	-0.133**
	(0.040)	(0.054)	(0.055)
No. of children $\geq$ 4	-0.220*	-0.177	-0.265*
	(0.113)	(0.114)	(0.156)
Boy	-0.006		
	(0.005)		
Gender Equality Test			
Diff in no. of children = 3		$p = 0.124$	
Diff in no. of children $\geq$ 4		$p = 0.612$	
Kleibergen-Paap F-statistic	1,471.555	413.138	1,293.723
Sample mean of the outcome variable	0.693	0.695	0.691
(Centered) R-squared	0.287	0.285	0.291
Observations	38,829	19,276	19,553

Notes: Data is from 2016 China Migrants Dynamic Survey (CMDS). To capture non-linear birth order profile in a non-linear family size profile setting, we follow Bagger et al. (2021) to include birth order indicators for the second, third, and fourth child or later. All columns additionally include controls containing destination province fixed effects, a quadratic in age (child, mother, and father), a dummy for whether a child is of primary school age, maternal age at second birth, dummies for maternal and paternal education attainment (middle school, high school, and college or above), dummies for migration distance (within province and within prefecture), indicator variables for whether both parents migrant, whether any grandparent migrants, whether the family is willing to settle down, and the duration of migration for the main parent respondent in years. In spirit of Angrist et al. (2010), we use the dummy for twinning at second parity and the full-sample dummy for twinning at third parity or above as the instruments. Standard errors clustered at the household level are in the parentheses.
***p<.01, **p<.05, *p<.10.

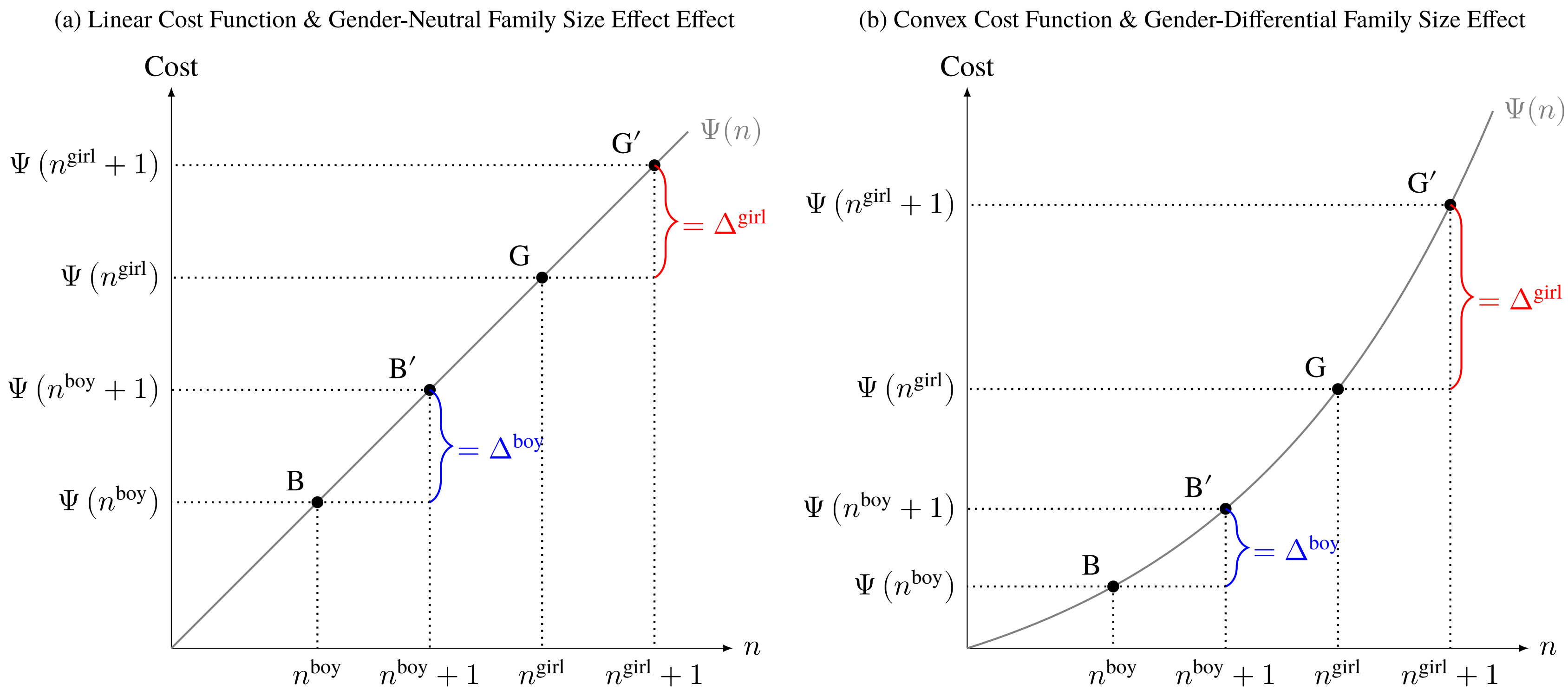


Figure A1: Linear *versus* Convex Cost Function of Migrating Children

Notes: The diagrams illustrate the differential impact of a convex versus linear cost function. Each cost function $\Psi(n)$ increases with family size n. The left panel depicts a linear increase, where marginal costs are constant. The right panel depicts a convex (accelerating) increase, where marginal costs rise with each additional child. Given son-biased fertility preferences, girls are overrepresented in *larger* families, while boys are overrepresented in *smaller* families. Consequently, girls are concentrated at the upper end and boys at the lower end of the family size (n) distribution. Let $\Delta \equiv \frac{\Psi(n+1)-\Psi(n)}{(n+1)-n}$ denote the marginal cost of increasing family size from n to $n+1$. In the linear case (left panel), given the same slope, the marginal cost of migrating an additional child is constant for boys and girls, i.e., $\Delta^{\text{boy}} = \Delta^{\text{girl}}$. However, the convex case (right panel) shows an increasing marginal cost, reflecting that migrating each additional child is progressively more expensive. Given that the slope now increases in family size, the marginal cost of migrating an additional child is increasingly higher for larger families compared to smaller counterparts. Thus, given girls' overrepresentation in larger families, the marginal cost of migrating is larger for girls than boys, i.e., $\Delta^{\text{boy}} < \Delta^{\text{girl}}$.

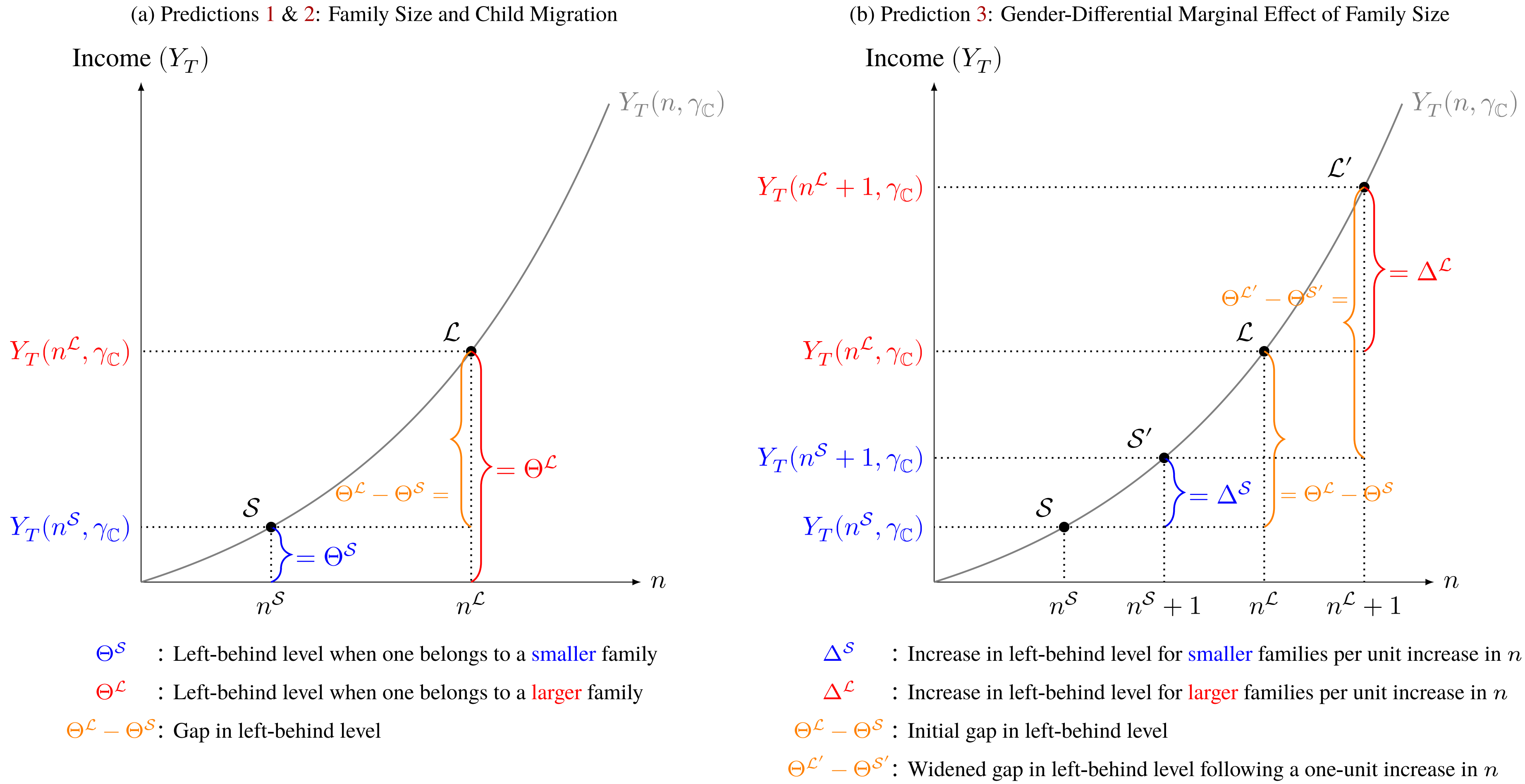
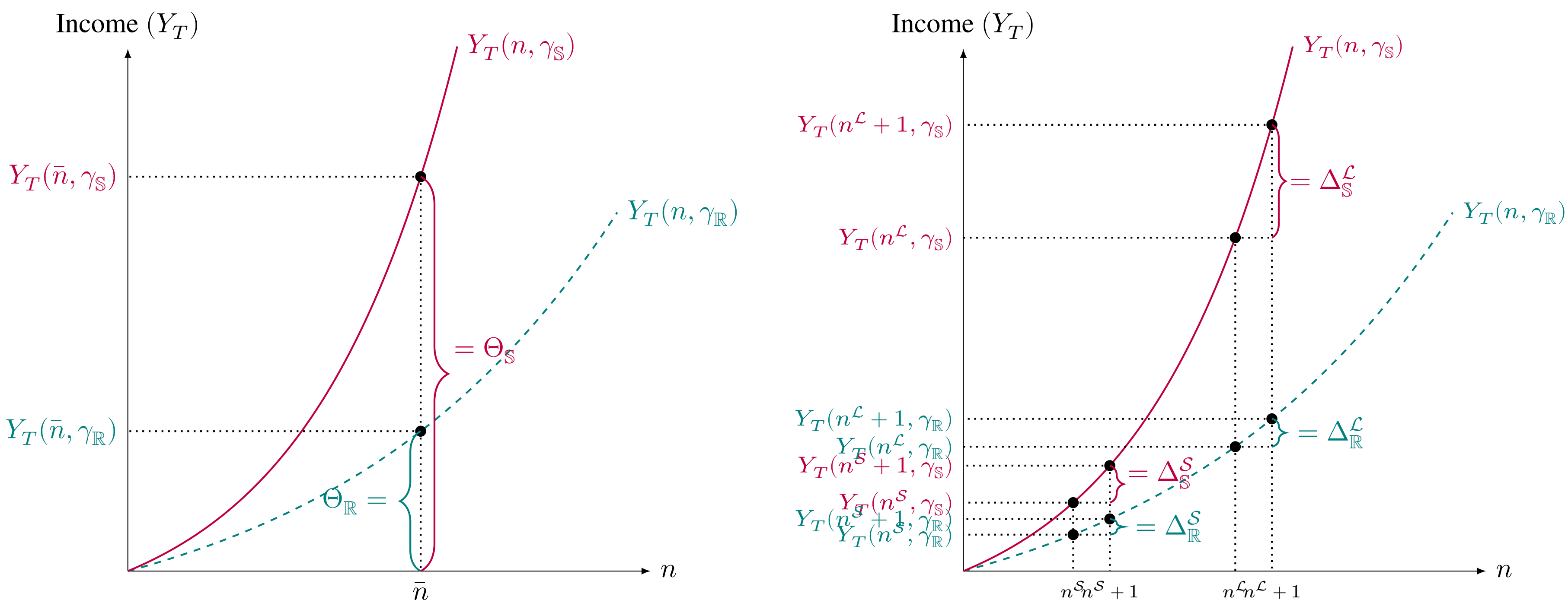


Figure A2: Main Model Predictions 1–3 — Threshold Incomes when Girls Have More Siblings than Boys

Notes: The figure illustrates the threshold incomes above which parents will migrate with children and below which parents will leave children behind. We define Θ as the level of households whose income is insufficient to exceed the threshold, leading them to leave all children behind. Thus, Θ represents the level of left-behind children. Δ denotes the marginal increase in the level of left-behind when family size increasing from n to $n+1$. $Y_T(n, \gamma_{\mathbb{C}})$ denotes the threshold for destination city C. $n^{\mathcal{S}}$ denotes the family size of children from smaller families (i.e., boys) and $n^{\mathcal{L}}$ denotes the family size of children from smaller families (i.e., girls).

(a) Prediction A1: Migration Restrictions and Child Migration

(b) Prediction A2: Migration Restrictions Amplifies Gender-Differential Family Size Effect



$\Theta_{\mathbb{R}}$: Left-behind when one's parents migrate to a relaxed city

$\Theta_{\mathbb{S}}$: Left-behind when one's parents migrate to a strict city

$\Delta^{\mathcal{L}}_{\mathbb{R}} - \Delta^{\mathcal{S}}_{\mathbb{R}}$: Larger vs. Smaller family: Gap in the marginal effect of n in a relaxed city

$\Delta^{\mathcal{L}}_{\mathbb{S}} - \Delta^{\mathcal{S}}_{\mathbb{S}}$: Larger vs. Smaller family: Gap in the marginal effect of n in a strict city

Figure A3: Main Model Predictions A1–A2 — Threshold Incomes when Girls Have More Siblings than Boys

Notes: The figure illustrates the threshold incomes above which parents will migrate with children and below which parents will leave children behind. We define Θ as the level of households whose income is insufficient to exceed the threshold, leading them to leave all children behind. Thus, Θ represents the level of left-behind children. Δ denotes the marginal increase in the level of left-behind when family size increasing from n to $n+1$. $Y_T(n, \gamma_{\mathbb{C}})$ denotes the threshold for destination city C. $n^{\mathcal{S}}$ denotes the family size of children from smaller families (i.e., boys) and $n^{\mathcal{L}}$ denotes the family size of children from smaller families (i.e., girls). $Y_T(n, \gamma_{\mathbb{S}})$ and $Y_T(n, \gamma_{\mathbb{R}})$ denote the thresholds for strict and relaxed cities, respectively.

(a) Marginal Treatment Effects for Boys

(b) Marginal Treatment Effects for Girls

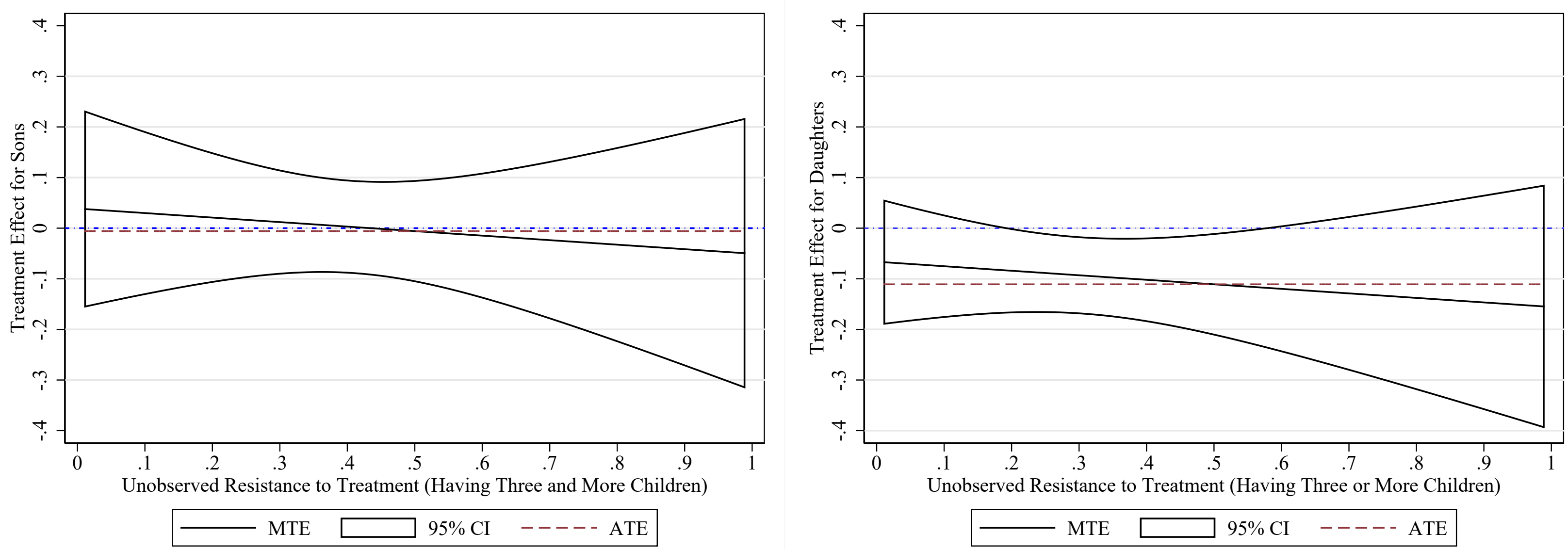


Figure A4: Marginal Treatment Effects

Notes: The figure plots the MTEs from the sub-samples of boys and girls. All the controls listed in Section 3 are included. The MTE estimation is based on a separate estimation procedure using a linear polynomial specification following Brinch et al. (2017). The horizontal axis in Panel (a) and (b) is the predicted probability of having three or more children after residualizing out all covariates including province fixed effects, that is, V_{ijp}^{D} in our notation. The vertical axis in Panel (a) and (b) is the MTE of having three and more children (rather than two) on the probability of child migration for boys or girls. Following Andresen (2018) and Mogstad and Torgovitsky (2018), we compute ATE as a weighted average of MTE. All estimations were done via the estimation program provided in Andresen (2018).

References

Andresen, Martin Eckhoff. 2018. "Exploring Marginal Treatment Effects: Flexible Estimation Using Stata." *Stata Journal* 18 (1): 118–158.

Angrist, Joshua D., Victor Lavy, and Analia Schlosser. 2010. "Multiple Experiments for the Causal Link between the Quantity and Quality of Children." *Journal of Labor Economics* 28 (4): 773–824.

Bagger, Jesper, Javier A Birchenall, Hani Mansour, and Sergio Urzúa. 2021. "Education, Birth Order and Family Size." *Economic Journal* 131 (633): 33–69.

Barcellos, Silvia Helena, Leandro S. Carvalho, and Adriana Lleras-Muney. 2014. "Child Gender and Parental Investments In India: Are Boys and Girls Treated Differently?" *American Economic Journal: Applied Economics* 6 (1): 157–189.

Basu, Deepankar, and Robert De Jong. 2010. "Son Targeting Fertility Behavior: Some Consequences and Determinants." *Demography* 47 (2): 521–536.

Bhalotra, Sonia, and Damian Clarke. 2020. "The Twin Instrument: Fertility and Human Capital Investment." *Journal of the European Economic Association* 18 (6): 3090–3139.

Bratti, Massimiliano, Simona Fiore, and Mariapia Mendola. 2020. "The Impact of Family Size and Sibling Structure on the Great Mexico–USA Migration." *Journal of Population Economics* 33 (2): 483–529.

Brinch, Christian N., Magne Mogstad, and Matthew Wiswall. 2017. "Beyond LATE with a Discrete Instrument." *Journal of Political Economy* 125 (4): 985–1039.

Chen, Yuanyuan, and Wei Fu. 2023. "Migration Control Policy and Parent–Child Separation Among Migrant Families: Evidence from China." *Journal of Population Economics* 36 (4): 2347–2388.

Epstein, Gil S. 2008. "Herd and Network Effects in Migration Decision-Making." *Journal of Ethnic and Migration Studies* 34 (4): 567–583.

Jensen, Robert. 2003. "Equal Treatement, Unequal Outcomes? Generating Sex Inequality Through Fertility Behavior." mimeo, Harvard University, Cambridge, MA.

Kuhn, Peter, and Kailing Shen. 2015. "Do Employers Prefer Migrant Workers? Evidence from a Chinese Job Board." *IZA Journal of Labor Economics* 4 (1): 22.

Mogstad, Magne, and Alexander Torgovitsky. 2018. "Identification and Extrapolation of Causal Effects with Instrumental Variables." *Annual Review of Economics* 10 (1): 577–613.

Nevo, Aviv, and Adam M Rosen. 2012. "Identification With Imperfect Instruments." *Review of Economics and Statistics* 94 (3): 659–671.

Oster, Emily. 2019. "Unobservable Selection and Coefficient Stability: Theory and Evidence." *Journal of Business & Economic Statistics* 37 (2): 187–204.

Peng, Yinni. 2019. "Getting Rural Migrant Children into School in South China: Migrant Agency and Parenting." *Asian Population Studies* 15 (2): 172–189.

Rosenzweig, Mark R., and Junsen Zhang. 2009. "Do Population Control Policies Induce More Human Capital Investment? Twins, Birth Weight and China's "One-Child" Policy." *Review of Economic Studies* 76 (3): 1149–1174.

Wang, Xiaobing, Yu Bai, Linxiu Zhang, and Scott Rozelle. 2017. "Migration, Schooling Choice, and Student Outcomes in China." *Population and Development Review* 43 (4): 625–643.

Wang, Zhiling, and Lu Chen. 2019. "Destination Choices of Chinese Rural–Urban Migrant Workers: Jobs, Amenities, and Local Spillovers." *Journal of Regional Science* 59 (3): 586–609.

Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. MIT Press.